System Identification and Parameter Estimation

Bin Yao

[1] School of Mechanical Engineering, Purdue University, West Lafayette, Indiana 47907-2088, USA. byao@purdue.edu
[2] Chang Jiang Chair Professor, The State Key Laboratory of Fluid Power and Mechatronic Systems, Zhejiang University, Hangzhou 310027, China

System identification deals with the problem of building mathematical models of dynamical systems based on observed data from the system. Thus it is the first step in any model-based feedback control design process. The subject is also of significant practical importance since the quality of resulting models has a heavy influence on the achievable closed-loop performance of the controlled system. This chapter first gives a brief overview of the general procedure of system identification and common issues to be addressed. The chapter then focuses on one of the key ingredients of system identification – parameter estimation. To make the development easy to understand, all algorithms will be developed in the discrete time domain first and their counterparts in continuous time domain will then be presented.

1 General Procedure and Common Issues

Consider a dynamic system illustrated in Fig. 1, in which $y, u, w : \mathcal{R} \to \mathcal{R}$ represent the system output function, the input function and/or measured disturbance function, and the unmeasured disturbance function, respectively. Such a dynamic system can in general be described by a mapping from input and disturbance function spaces to the output function space as

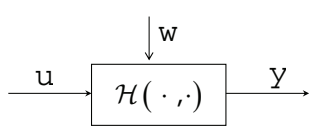

Fig. 1. A dynamic system with output y, input and/or measured disturbance u, and unmeasured disturbance w

$$y = \mathcal{H}(u, w), \quad u \in \Omega_u,\ w \in \Omega_w,\ \mathcal{H} : \mathcal{L}_{\infty e} \times \mathcal{L}_{\infty e} \to \mathcal{L}_{\infty e} \tag{1}$$

where $\Omega_u \subset \mathcal{L}_{\infty e}$ and $\Omega_w \subset \mathcal{L}_{\infty e}$ represent the set of all possible functions of u and w in reality, respectively. System identification is the process to obtain the mapping $\mathcal{H}$ based on experimental data collected through conducting certain experiments. Specifically, let $\mathcal{U}_t$ and $\mathcal{Y}_t$ be the data sets of all input/measured-disturbance u and output y collected by the sampling instance t respectively. The system identification problem is to construct an estimate of $\mathcal{H}$, $\hat{\mathcal{H}}$, based on the collected experimental data set $\mathcal{Z}_t = \{\mathcal{U}_t,\ \mathcal{Y}_t\}$ at time t, in which $\mathcal{U}_t = \{u(t), u(t-1), \ldots\}$ and $\mathcal{Y}_t = \{y(t), y(t-1), \ldots\}$. Thus, a system identification consists of the following elements and issues to be addressed:

1.1 *Model Structure Selection*

The overall purpose of system identification is to produce a compact mathematical model that can be stored and used easily and yet capable of adequately describing the actual system's behavior during its intended use. As such, it is necessary to restrict the model to a set of known models or structures which can be conveniently stored and/or easily analyzed later. For simplicity of notation, assume that the set of candidate models or structures is parametrized by $\boldsymbol{\theta}$, a vector of finite dimension p within a known set $\Omega_\theta \subset \mathcal{R}^p$ as

$$\mathcal{M} := \{\mathcal{M}(\boldsymbol{\theta}) :\ \mathcal{M}(\boldsymbol{\theta}) = \mathcal{G}_\theta(u, w),\ \boldsymbol{\theta} \in \Omega_\theta\} \tag{2}$$

where $\mathcal{G}_\theta$ is a known model depending on $\boldsymbol{\theta}$. The selection of a suitable model structure for a particular system identification is in general influenced by the following several factors:

F1. *The ultimate purpose of a system identification.*
Various system identifications have slightly different goals in mind. For some applications, such as realistic simulations of physical systems to reduce the development time and cost of producing new products, the purpose of system identification is to produce a model that captures all behaviors of the actual system as accurately as possible. For these applications, the model structure (2) has to be chosen large and complex enough to have the ability to describe various system behaviors precisely (e.g., the nonlinear Coulomb friction in mechanical systems can never be adequately represented by linear models, thus the model structure has to include nonlinearity to be meaningful). However, the model structure should not be so overly large and complex that either the certain identifiability condition (e.g., the invertibility of the model structure introduced in [6]) is lost and no particular model can be determined based on all available measurement data or the resulting model is so complicated that it is impossible or difficult to use later. Thus, a trade-off usually has to be made between the model complexity needed for accurate representation of all actual system

behaviors and simplicity for practical use. In this regard, the first principle based conventional physical modeling becomes very helpful. This is especially true for system identification of mechanical systems which can incorporate modeling knowledge from related disciplines (e.g., mechanics and thermodynamics).

For some applications such as system identification for model-based controller designs, the purpose is not to truthfully duplicate all system behaviors but to precisely capture the characteristics of the actual system that are essential to the subsequent controller design and/or have limiting effects on the achievable closed-loop performance. For these applications, the selection of appropriate model structure is determined by the controller design methodology to be used (e.g., the traditional H_∞ robust control demands a linear time-invariant (LTI) model with certain known bounds on modeling uncertainties) as well as the need for simplicity that allows the resulting identified model to be easily used for subsequent controller design, analysis, and implementation, in which on-line computation cost is a major concern.

F2. *The measurable/available experimental data.*
Given any chosen model structure (2), the exact model parameters of a system identification for $\boldsymbol{\theta}$ will have to be determined based on the actual experimental data. For some applications, this data set may never be so rich that all phenomena or model parameters of interests can be determined (e.g., the on-board modeling and identification studied in [**?**]). For these applications, though it might be at the expense of losing some generality and accuracy of the identification results, it may be necessary to further constrain the model structure to be used to make sure that certain model parameter identifiability conditions are satisfied under the constraints of available experimental data in practices. For other applications, one may run into situations where after various trials no model parameters can be found within the given model structure to adequately capture all the important characteristics shown in the data; it might well be that the chosen model structure is too small and needs to be enlarged instead.

Model structure selection is an important task to perform as it is directly tied to the ultimate purpose of a system identification. Unfortunately, it is also one of the most difficult questions to answer since it is very hard, if not impossible, to quantify various factors influencing the selection of a suitable model structure. Physical insights into actual system dynamics and good knowledge of various model structures and their properties (e.g., the phenomena which can be described by the model structure, the parameter identifiability condition, ...) play a significant role in addressing this problem well. Some of the well-studied model structures for linear time-invariant (LTI) systems in discrete-time domain are:

M1. *Auto Regressive with Exogenous Input (ARX) Models*

The ARX model structure is described by the following LTI difference equation:

$$\begin{aligned} y(t) + a_1 y(t-1) + \ldots + a_{n_a} y(t-n_a) \\ = b_0 u(t-d) + b_1 u(t-d-1) + \ldots + b_{n_b} u(t-d-n_b) \end{aligned} \tag{3}$$

in which output of the system at instance t is a weighted average of the past outputs over a horizon of n_a samples (i.e., Auto Regressive) plus a weighted average of inputs over a horizon of n_b samples delayed by d samples (i.e., with Exogenous Input). For simplicity, let $n = max\{n_a,\ n_b + d\}$ be the order of the system and define

$$\begin{aligned} A(q^{-1}) &= 1 + a_1 q^{-1} + \ldots + a_{n_a} q^{-n_a} \\ B(q^{-1}) &= b_0 + b_1 q^{-1} + \ldots + b_{n_b} q^{-n_b} \end{aligned} \tag{4}$$

where q^{-1} represents the backward shift operator (e.g., $q^{-1}y(t) = y(t-1)$). With these notations, (3) can be put in a short form as

$$A(q^{-1})y(t) = q^{-d}B(q^{-1})u(t), \quad \text{or} \quad y(t) = G(q^{-1})u(t) \tag{5}$$

where

$$G(q^{-1}) = \frac{q^{-d}B(q^{-1})}{A(q^{-1})} \tag{6}$$

is called the *transfer operator* [5]. Note that with z^{-1} for q^{-1}, $H(z) = G(z^{-1})$ becomes the usual transfer function of the LTI discrete-time system (3). However, it should be noted that the transfer operator (6) can be defined even if the parameters of the model (3) are time-varying, while the concept of transfer function is only defined for LTI systems and does not exist at all when parameters are time-varying. Note that for any given set of n_a, n_b and d, the ARX model structure can be linearly parametrized by a vector of parameters $\boldsymbol{\theta}$ as

$$y(t) = \boldsymbol{\varphi}(t)^T \boldsymbol{\theta} \tag{7}$$

where

$$\begin{aligned} \boldsymbol{\varphi}(t)^T &= [-y(t-1), \ldots, -y(t-n_a),\ u(t-d), \ldots, u(t-d-n_b)] \\ \boldsymbol{\theta} &= [a_1,\ \ldots,\ a_{n_a},\ b_0,\ \ldots, b_{n_b}]^T \in R^{n_a+n_b+1} \end{aligned} \tag{8}$$

M2. *ARMAX, Output-Error (OE), and Box-Jenkins (BJ) Models*

The ARX model structure does not take disturbances into account. For systems with significant disturbance effect, it might be beneficial to explicitly model the disturbance and its effect as:

$$A(q^{-1})y(t) = \tfrac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t) + \tfrac{C(q^{-1})}{D(q^{-1})}d_e(t) \tag{9}$$

where $L(q^{-1}) = 1+l_1q^{-1}+\ldots+l_{n_l}q^{-n_l}$, $C(q^{-1}) = 1+c_1q^{-1}+\ldots+c_{n_c}q^{-n_c}$, $D(q^{-1}) = 1 + d_1q^{-1} + \ldots + d_{n_d}q^{-n_d}$, and $d_e(t)$ is normally assumed to be a zero mean white noise sequence. In (9), $H_u(z) = \frac{z^{-d}B(z^{-1})}{A(z^{-1})L(z^{-1})}$ and $H_e(z) = \frac{C(z^{-1})}{A(z^{-1})D(z^{-1})}$ can be thought of as the transfer functions from u and d_e to y respectively; $H_e(z)$ represents the lumped disturbance model, and $A(z^{-1})$ captures all the common poles of the two transfer functions.

The structure (9) is quite general. Most often the choices are confined to one of the special cases:
ARX

$$A(q^{-1})y(t) = q^{-d}B(q^{-1})u(t) + d_e(t) \tag{10}$$

ARMAX

$$A(q^{-1})y(t) = q^{-d}B(q^{-1})u(t) + C(q^{-1})d_e(t) \tag{11}$$

OE:

$$y(t) = \frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t) + d_e(t) \tag{12}$$

BJ :

$$y(t) = \frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t) + \frac{C(q^{-1})}{D(q^{-1})}d_e(t) \tag{13}$$

It is clear that once the orders of all the polynomials are specified, all the above model structures can be parametrized by a vector of parameters consisting of coefficients of all the polynomials.

1.2 *Criterion of Fit or Specific Estimation Method*

Once the targeted model structure set (2) is specified, the immediate question arises of how to select a particular model from the set using collected experimental data $\mathcal{Z}_t$. For this purpose, a criterion of fit or a specific parameter estimation algorithm needs to be specified to compute the "best" model in the model structure for a good fit of the date set $\mathcal{Z}_t$.

A natural choice for the criterion of fit is to minimize the discrepancy between the measured output of a system and the predicted output of the model using the same set of past input and output measurement data. As the model structure may contain some random signals (e.g., the zero mean white noise sequence $d_e(t)$ in (9)), the predicted output of the model using the same set of past input and output measurement data cannot be precisely determined in reality. To solve this ambiguity, the optimal predictor of a model structure based on the past measurement data will be used to fit the measured output of the actual system instead. Specifically, a $j \leq d$ step ahead predictor based on the information available up to the time instance t is a function of the collected experimental data set $\mathcal{Z}_t$:

$$\hat{y}(t+j \mid t) = f[\boldsymbol{\theta}, y(t), y(t-1), \ldots, u(t), u(t-1), \ldots] \tag{14}$$

With the least-squares-estimation (LSE), the performance index would be to minimize the following index representing the discrepancy between the measured output and the predicted output of the model with the same set of past input and measured output for all time up to t:

$$\begin{aligned} J(t,\boldsymbol{\theta}) &= \textstyle\sum_{i=1}^{t-j} \left[y(i+j) - \hat{y}(i+j \mid i)\right]^2 \\ &= \textstyle\sum_{i=1}^{t-j} \left[y(i+j) - f[\boldsymbol{\theta}, y(t), y(t-1), \ldots, u(t), u(t-1), \ldots]\right]^2 \end{aligned} \tag{15}$$

in which $y(i+j)$ is the measured actual output at time $i+j$. The "best" identified model parameter at time t can then be obtained by solving the following nonlinear optimization problem:

$$\hat{\boldsymbol{\theta}}(t) = \arg\left\{\min_{\boldsymbol{\theta}\in\Omega_\theta} J(t,\boldsymbol{\theta})\right\} \tag{16}$$

For example, the j step ahead optimal predictor of a model structure like (9) is a predictor of the form (14) that minimizes

$$E\left\{[y_m(t+j) - \hat{y}(t+j \mid t)]^2\right\} \tag{17}$$

where E represents the expectation operator and $y_m(t+j)$ is the random output of the model structure (9) at instance $t+j$ with the same past input and output data set $\mathcal{Z}_t$ and the assumed random disturbances $d_e(t)$.

Lemma 1. *Assuming that $C(q^{-1})$ is asymptotically stable with $n_c \le n_a + j - 1$ and $d_e(t)$ is a zero mean white noise, the optimal j step ahead predictor of the ARMAX model (11) is given by*

$$\hat{y}(t+j \mid t) = \tfrac{F_j(q^{-1})}{C(q^{-1})} y(t) + \tfrac{J(q^{-1})B(q^{-1})}{C(q^{-1})} u(t-d+j) \tag{18}$$

where $F_j(q^{-1}) = f_{j0} + f_{j1}q^{-1} + \ldots + f_{jn_f}q^{-n_f}$ with $n_f = n_a - 1$ and $J(q^{-1}) = 1 + j_1 q^{-1} + \ldots + j_{j-1}q^{-j+1}$ are solutions to the following polynomial equation:

$$C(q^{-1}) = A(q^{-1})J(q^{-1}) + q^{-j}F_j(q^{-1}) \tag{19}$$

△

Denote

$$G_j(q^{-1}) = J(q^{-1})B(q^{-1}) = g_0 + g_1 q^{-1} + \ldots + g_{n_b+j-1}q^{-(n_b+j-1)} \tag{20}$$

The optimal j step ahead predictor (18) can then be put into the following *pseudolinear regression* form:

$$\hat{y}(t+j \mid t) = \boldsymbol{\varphi}(t,\boldsymbol{\theta})^T \boldsymbol{\theta} \tag{21}$$

with

$$\begin{aligned} \boldsymbol{\theta} &= [f_{j0}, \ldots, f_{jn_f},\ g_0, \ldots, g_{n_b+j-1},\ c_1, \ldots, c_{n_c}]^T \\ \boldsymbol{\varphi}(t,\boldsymbol{\theta})^T &= [y(t), \ldots, y(t-n_f),\ u(t-d+j), \ldots, u(t-d-n_b+1), \\ &\qquad -\hat{y}(t-1+j \mid t-1), \ldots, -\hat{y}(t-n_c+j \mid t-n_c)] \end{aligned} \tag{22}$$

in which $\boldsymbol{\varphi}$ is a function of $\boldsymbol{\theta}$ as well due to the fact that the past optimal predicted output $\hat{y}(t+j-k \mid t-k)$ has to be calculated from the collected experimental data set $\mathcal{Z}_{t-k}$ from (18) which depends on $\boldsymbol{\theta}$. With this optimal j step ahead predictor (21), the "best" identified parameters of the ARMAX model (11) can then be obtained using the LSE algorithm (16).

It is thus clear that even with the simple LTI ARMAX model structure (11), the performance index (15) is likely a nonlinear function of $\boldsymbol{\theta}$ and the model parameters have to be solved numerically via certain iteration algorithms in general. It is also a high possibility that local minimum solutions may result unless certain priori knowledge is incorporated into the selection of initial values for iteration.

1.3 *Experimental Design and Data Collection and Processing*

Any system identification starts with dedicated experimental design and data collection. This as well as the subsequent data processing before the data are passed to any parameter estimation algorithm may have significant impact on the quality of the identified model parameters due to the following two main factors. First, for a given model structure, the data set has to be informative enough (captured by certain persistent exciting conditions introduced later) to be able to uniquely determine model parameters based on the experimental data. This in practice can be achieved only through well thought-out experiments. Secondly, there will always exist some mismatch between the actual system dynamics and the chosen model structure (i.e., model error) no matter how complex a model structure is. Even with the same physical system the characteristics of the model error depends on the chosen model structure. Thus, how to select a model structure which minimizes overall model error and model complexity is one of the major concerns. In addition, after a suitable model structure is determined, actual values of the model error still depend on the experimental data used and different sets of experimental data may lead to significant discrepancies among identified model parameters. Thus, once a model structure is chosen and the characteristics of the model error is known, how to process the collected experimental data record (e.g., remove bad data and offsets, proper filtering [10] and segmentation of data) to obtain a date set which would minimize the effect of the model error on the consistency and accuracy of the resulting identified model parameters becomes an important practical issue to deal with as well.

1.4 *Model Validation*

Once the model parameters are determined, the next task is model validation which asks if the identified model is indeed useful. Model validation is a highly subjective task that scrutinizes all properties of the model to see if the model serves the need of the end user. In practice, comparing the predicted model

output with measured output that was not used for the fit (called validation data) can test the model's ability to reproduce the behavior of the validation data set by simulation and prediction. It is also valuable to look at the data that could not be reproduced by the model (the residuals). If an identified model captures all relevant actual system dynamics, the residuals should not be correlated with other available information, such as the system's input.

The overall system identification procedure is shown in Fig. 2. A good system identification normally undergoes several iterations since various steps are intertwined and some of them are quite subjective. Though various existing identification techniques and the rationales behind them can be taught along with numerical schemes in computing the estimates in a book, due to the subjectivity of experimental design, choosing model structure and identification criterion, and processing and selecting suitable date sets for parameter estimation, the success of a system identification in practice still heavily depends on the user's understanding of the subject. In this regard, knowing the properties of various models and estimation algorithms is the key for a user to make full use of prior information and the observed data in coming out intelligent choices of experimental design, suitable model structure, specific estimation algorithm, and data processing. For an in-depth treatment of the subject, the reader is referred to [6].

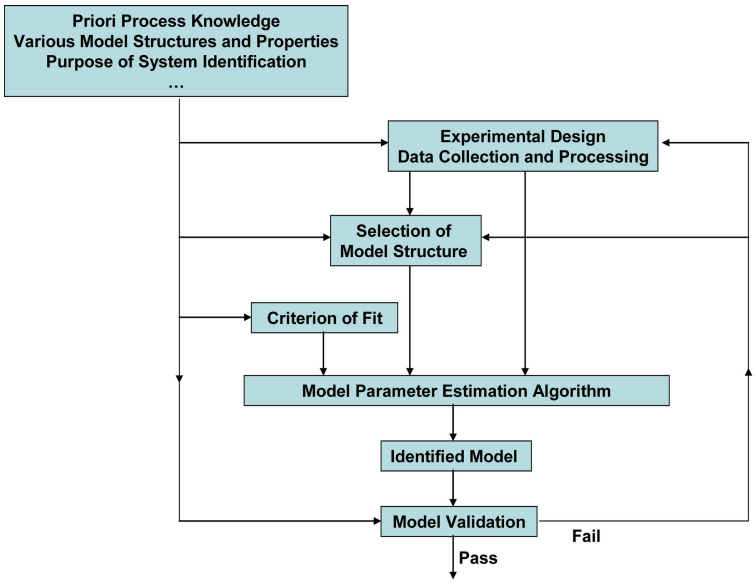

Fig. 2. The system identification procedure

2 Parameter Estimation for Linear Regression Type Model Structures

In this section, the scope is restricted to parameter estimation for model structures in which certain prediction output can be linearly parametrized by the model parameters through re-grouping and transformation if necessary. Specifically, consider the following *linear regression* form:

$$\hat{\mathbf{y}}(t,\boldsymbol{\theta}) = \boldsymbol{\varphi}_1(t)\theta_1 + \boldsymbol{\varphi}_2(t)\theta_2 + \cdots + \boldsymbol{\varphi}_p(t)\theta_p = \boldsymbol{\varphi}(t)^T\boldsymbol{\theta} \tag{23}$$

where $\hat{\mathbf{y}} \in \mathcal{R}^m$ represents the predicted output vector of the model with parameter vector $\boldsymbol{\theta}$ at time t, in which m is the number of the outputs. Furthermore, $\boldsymbol{\varphi}_i(t) \in \mathcal{R}^m$ is the vector of some known basis functions's values at t corresponding to θ_i; these values can be nonlinear functions of the system state so long as they can be calculated once the experimental data set $\mathcal{Z}_t$ is given. $\boldsymbol{\varphi}(t) = \left[\boldsymbol{\varphi}_1(t)\ \boldsymbol{\varphi}_2(t)\ ...\ \boldsymbol{\varphi}_p(t)\right]^T \in \mathcal{R}^{p\times m}$ is normally referred to as the regressor matrix at t. Finally, $\boldsymbol{\theta} = [\theta_1, ..., \theta_p]^T \in \mathcal{R}^p$ is the vector of p unknown model parameters that we would like to determine so that the predicted output $\hat{\mathbf{y}}$ overall fits the measured output $\mathbf{y}$ well in certain sense.

2.1 Least Square Estimation (LSE) Algorithm

Least square estimation (LSE) is perhaps the most popular off-line parameter estimation method [4] for linear regression model structures. The goal is to determine model parameters $\boldsymbol{\theta}$ in (23) so that the predicted output $\hat{\mathbf{y}}$ overall fits the measured output $\mathbf{y}$ well as measured by minimizing the following summation of weighted fitting-error-squares type performance index function

$$\begin{aligned} J(t,\boldsymbol{\theta}) &= \sum_{i=1}^{t} \left\|\mathbf{R}(t,i)\left[\mathbf{y}(i) - \boldsymbol{\varphi}(i)^T\boldsymbol{\theta}\right]\right\|^2 = \sum_{i=1}^{t} \boldsymbol{\epsilon}_\theta(i)^T \underbrace{\mathbf{R}(t,i)^T\mathbf{R}(t,i)}_{\mathbf{W}(t,i)} \boldsymbol{\epsilon}_\theta(i) \\ &= \mathbf{E}_\theta(t)^T\bar{\mathbf{W}}(t)\mathbf{E}_\theta(t), \qquad \bar{\mathbf{W}}(t) = diag\left\{\mathbf{W}(t,1), \ldots \mathbf{W}(t,t)\right\} \end{aligned} \tag{24}$$

where $\mathbf{R}(t,i)$ or $\mathbf{W}(t,i)$ represents the weighting matrix, $\boldsymbol{\epsilon}_\theta(i) = \mathbf{y}(i) - \hat{\mathbf{y}}(i,\boldsymbol{\theta})$ is the model prediction error at time i with model parameter vector $\boldsymbol{\theta}$, and $\mathbf{E}_\theta = \left[\boldsymbol{\epsilon}_\theta(1)^T, \boldsymbol{\epsilon}_\theta(2)^T, \cdots, \boldsymbol{\epsilon}_\theta(t)^T\right]^T$ represents the whole model error vector. For simplicity, denote

$$\mathbf{Y}(t) = \begin{pmatrix} \mathbf{y}(1) \\ \vdots \\ \mathbf{y}(t) \end{pmatrix} \in R^{tm}, \quad \boldsymbol{\Phi}(t) = \begin{pmatrix} \boldsymbol{\varphi}(1)^T \\ \vdots \\ \boldsymbol{\varphi}(t)^T \end{pmatrix} \in R^{tm\times p} \tag{25}$$

Then

$$\mathbf{E}_\theta(t) = \mathbf{Y}(t) - \boldsymbol{\Phi}(t)\boldsymbol{\theta} \tag{26}$$

The solution to the above minimization problem or the LSE of $\hat{\boldsymbol{\theta}}(t)$ can be found through the necessary condition for optimality:

$$\left.\frac{\partial J(t,\boldsymbol{\theta})}{\partial \boldsymbol{\theta}}\right|_{\boldsymbol{\theta}=\hat{\boldsymbol{\theta}}(t)} = \mathbf{0} \tag{27}$$

which leads to the following normal equation [1, 5]:

$$\boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \boldsymbol{\Phi}(t) \hat{\boldsymbol{\theta}}(t) = \boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \mathbf{Y}(t) \tag{28}$$

Thus, when $\boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \boldsymbol{\Phi}(t)$ is nonsingular, the minimization solution is unique and the resulting LSE can be calculated by

$$\hat{\boldsymbol{\theta}}(t) = \mathbf{P}(t) \boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \mathbf{Y}(t) = \mathbf{P}(t) \left[\sum_{i=1}^{t} \boldsymbol{\varphi}(i) \mathbf{W}(t,i) \mathbf{y}(i) \right] \tag{29}$$

where

$$\mathbf{P}(t) = \left[\boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \boldsymbol{\Phi}(t) \right]^{-1} = \left[\sum_{i=1}^{t} \boldsymbol{\varphi}(i) \mathbf{W}(t,i) \boldsymbol{\varphi}(i)^T \right]^{-1} \in \mathcal{R}^{p \times p} \tag{30}$$

which is normally referred to as the *covariance matrix*.

2.2 Convergence of LSE under no Model Structure Error

To see how well the LSE result (29) performs, the ideal case where the measured output $\mathbf{y}(t)$ is generated from a system having the same dynamic model structure as the assumed one (23) (i.e., no model structure error) will be examined first. Namely, assume $\mathbf{y}(i) = \boldsymbol{\varphi}(i)^T \boldsymbol{\theta}^0$, $\forall i$, where $\boldsymbol{\theta}^0$ represents the true model parameters. In such a case, from (25), $\mathbf{Y}(t) = \boldsymbol{\Phi}(t)\boldsymbol{\theta}^0$, and from (29), $\hat{\boldsymbol{\theta}}(t) = \boldsymbol{\theta}^0$, which implies that the true model parameters can be accurately recovered by LSE method as long as $\boldsymbol{\Phi}(t)^T \bar{\mathbf{W}}(t) \boldsymbol{\Phi}(t)$ is nonsingular. A necessary condition for that to happen is $\boldsymbol{\Phi}(t)$ has a full column rank p. This leads to the following minimum conditions that the experimental design and data processing have to satisfy:

E1 . *The data set has to be large enough.* For single output model ($m = 1$), the number of data sampling instances has to be no less than p (i.e., $t \geq p$) for $\boldsymbol{\Phi}(t) \in \mathcal{R}^{t \times p}$ to have a rank of p.

E2 . *The data set has to be rich enough.* All columns of $\boldsymbol{\Phi}(t)$ have to be linearly independent for it to have a rank of p. Thus, it is important to remove segments of experimental data in which columns of $\boldsymbol{\Phi}(t)$ are strongly correlated with each other.

2.3 Effect of Model Error

In reality, there always exists certain amount of model error and it is important to examine how the model error affects accuracy of the resulting LSE estimates to look for practical ways to minimize its effect. For this purpose, let us assume that the actual process is governed by

$$\mathbf{y}(t) = \boldsymbol{\varphi}(t)^T \boldsymbol{\theta}^0 + \mathbf{e}(t) \tag{31}$$

where $\mathbf{e}(t)$ represents the model error. For this system, the LSE (29) becomes (for simplicity, assume $\mathbf{W}(t,i) = \mathbf{I}_p$)

$$\hat{\boldsymbol{\theta}}(t) = \boldsymbol{\theta}^0 + \mathbf{P}(t) \left[\sum_{i=1}^{t} \boldsymbol{\varphi}(i)\mathbf{e}(i) \right] \tag{32}$$

For any vector of signals $\{\mathbf{s}(t)\}$ which could be some random processes[3], define an expectation-like operation symbol $\bar{E}$ by [6]

$$\bar{E}[\mathbf{s}(t)] = \lim_{N\to\infty} \frac{1}{N} \sum_{t=1}^{N} E[\mathbf{s}(t)] \tag{33}$$

in which the expectation E is with respect to the "stochastic components" of $\mathbf{s}(t)$, i.e., if $\mathbf{s}(t)$ is a deterministic sequence, the expectation is without effect. Introduce the covariance-function-like quantity $\mathbf{R}_{\mathbf{s}}(\tau)$ as

$$\mathbf{R}_{\mathbf{s}}(\tau) := \bar{E}[\mathbf{s}(t)\mathbf{s}(t-\tau)^T] \tag{34}$$

With a large data set (i.e., when $t \to \infty$), from (32), the LSE estimation error $\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}^0$ approaches

$$\tilde{\boldsymbol{\theta}}(t) = \left(\tfrac{1}{t} \sum_{i=1}^{t} \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T \right)^{-1} \left(\tfrac{1}{t} \sum_{i=1}^{t} \boldsymbol{\varphi}(i)\mathbf{e}(i) \right) \;\to\; \mathbf{R}_{\boldsymbol{\varphi}}(0)^{-1}\mathbf{R}_{\boldsymbol{\varphi}\mathbf{e}}(0) \tag{35}$$

where $R_{\boldsymbol{\varphi}\mathbf{e}}(\tau) := \bar{E}[\boldsymbol{\varphi}(t)\mathbf{e}(t-\tau)]$, $\forall\tau$. Without loss of generality, assume $\bar{E}[\mathbf{e}(t)] = 0$ since the model structure (23) can always be augmented by a constant to remove the non-zero mean effect of the model error. (35) indicates that the parameter estimation accuracy can be improved through the following two practical means:

E3 . *Un-correlate the regressor vector $\boldsymbol{\varphi}(i)$ and the model error vector $\mathbf{e}(i)$.* Uncorrelated $\boldsymbol{\varphi}(i)$ and $\mathbf{e}(i)$ with $\bar{E}[\mathbf{e}(t)] = 0$ leads to $\mathbf{R}_{\boldsymbol{\varphi}\mathbf{e}}(0) = 0$. Thus, an unbiased estimate (i.e., $\tilde{\boldsymbol{\theta}}(t) \to 0$ as $t \to \infty$) is obtained as long as $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ is nonsingular, commonly referred to as the *persistent exciting (PE)* condition.

[3] refer Appendix A for an introduction to probability and random process

E4 . *Maximize the "signal to noise ratio" (SNR).* $\boldsymbol{\varphi}(t)^T\boldsymbol{\theta}^0$ is the output due to the parameters to be estimated (i.e., useful signal) and $\mathbf{e}(t)$ can be thought as noise. With $\bar{E}\left[\|\boldsymbol{\varphi}(t)^T\boldsymbol{\theta}^0\|^2\right] = \boldsymbol{\theta}^{0T}\mathbf{R}_{\boldsymbol{\varphi}}(0)\boldsymbol{\theta}^0 \geq \lambda_{min}\left[\mathbf{R}_{\boldsymbol{\varphi}}(0)\right]\|\boldsymbol{\theta}^0\|^2$ and $\bar{E}\left[\|\mathbf{e}(t)\|^2\right] = Tr\{\mathbf{R}_{\mathbf{e}}(0)\} \leq m\lambda_{max}\left[\mathbf{R}_{\mathbf{e}}(0)\right]$, $\lambda_{min}\left[\mathbf{R}_{\boldsymbol{\varphi}}(0)\right]/\lambda_{max}\left[\mathbf{R}_{\mathbf{e}}(0)\right]$ may be thought of as a measure of SNR. By designing experiments and processing data sets such that SNR is maximized, the LSE estimation error (35) can be reduced as well, even when $\boldsymbol{\varphi}(i)$ and $\mathbf{e}(i)$ may not be made uncorrelated. So it is important to look for practical ways to maximize the SNR like using as large inputs as possible to increase $\boldsymbol{\varphi}(t)$ while at the same time making sure that $\mathbf{e}(t)$ is kept small by not significantly violating the assumptions of the chosen model structure. Proper filtering input and output data to remove the effects of major disturbances as done in [10] is another convenient way to accomplish this objective.

2.4 Converging Rate

The above analysis shows that, even though the LSE estimation error is generally non-zero in the presence of model error, it can still be made to converge to zero as $t \to \infty$ when the regressor vector $\boldsymbol{\varphi}(i)$ and the model error vector $\mathbf{e}(i)$ are uncorrelated. In practice, the converging rate is also of significant importance since all practical experiments will have to be run within finite time periods. To gain some insights on converging rate of the LSE algorithm, the covariance of the LSE estimation error needs to be carefully examined. For simplicity, assume that the sequence of the regressors $\{\boldsymbol{\varphi}(t)\}$ is a deterministic sequence and the model error $\{\mathbf{e}(t)\}$ is a sequence of independent random variable vectors with zero mean values and covariances of $\sigma^2 I_m$. With these assumptions, $\boldsymbol{\varphi}(i)$ and $\mathbf{e}(i)$ are uncorrelated and $E\left\{\mathbf{e}(i)\mathbf{e}(j)^T\right\} = \sigma^2\delta_{ij}I_m$, $\forall i, j$. The covariance matrix of the LSE estimation error (32) thus becomes

$$\begin{aligned} cov\left(\tilde{\boldsymbol{\theta}}(t)\right) &= E\left\{\tilde{\boldsymbol{\theta}}(t)\tilde{\boldsymbol{\theta}}(t)^T\right\} = E\left\{\mathbf{P}(t)\left[\sum_{i=1}^{t}\sum_{j=1}^{t}\boldsymbol{\varphi}(i)\mathbf{e}(i)\mathbf{e}(j)^T\boldsymbol{\varphi}(j)^T\right]\mathbf{P}(t)^T\right\} \\ &= \mathbf{P}(t)\left[\sum_{i=1}^{t}\sum_{j=1}^{t}\boldsymbol{\varphi}(i)\sigma^2\delta_{ij}\boldsymbol{\varphi}(j)^T\right]\mathbf{P}(t) = \sigma^2\mathbf{P}(t) = \tfrac{1}{t}\sigma^2\left[t\mathbf{P}(t)\right] \\ &= \tfrac{1}{t}\sigma^2\left[\tfrac{1}{t}\sum_{i=1}^{t}\boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T\right]^{-1} \overset{\text{as } t\to\infty}{\longrightarrow} \tfrac{1}{t}\sigma^2\mathbf{R}_{\boldsymbol{\varphi}}(0)^{-1} \end{aligned} \tag{36}$$

(36) shows that, when the PE condition is satisfied, i.e., $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ is nonsingular or $\mathbf{R}_{\boldsymbol{\varphi}}(0) > 0$, the covariance matrix of the LSE estimation error converges to zero at a rate of $1/t$. The larger $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ is or the higher the signal excitation level is (as $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ depends on the regressor only), the more accurate the resulting LSE parameter estimates are.

Example 1. To better understand the effect of experimental conditions on the convergence rate and quality of resulting LSE estimates, let us consider the

estimation of a single parameter, i.e., consider the linear regression model $\hat{y}(t,\theta) = \varphi(t)\theta$ with an actual process of $y(t) = \varphi(t)\theta^0 + e(t)$. In such a case, $\mathbf{\Phi}(t)^T\mathbf{\Phi}(t) = \sum_{i=1}^{i=t} \varphi(i)^2$, which leads to

$$cov\left(\tilde{\theta}(t)\right) = \frac{\sigma^2}{\sum_{i=1}^{i=t} \varphi(i)^2} \tag{37}$$

If $\sum_{i=1}^{i=t} \varphi(i)^2 \to \infty$ as $t \to \infty$ (e.g., $|\varphi(i)| \geq$ any small constant), then $cov\left(\tilde{\theta}(t)\right) \to 0$, which means that the LSE $\hat{\theta}(t)$ converges to its true value θ^0 in spite of the zero mean white noise $e(t)$. It is also observed that

(1) . If $\varphi(i) \sim e^{-ai}, a > 0$, then, the variance goes to a constant and the parameter estimate does not converge.
(2) . If $\varphi(i) \sim 1$, then, the variance converges to zero at a rate of $1/t$.
(3) . If $\varphi(i) \sim e^{ai}, a > 0$, then, the variance converges to zero at an exponential rate of e^{-2at}.

The above cases illustrate that the converging rate of parameter estimation is affected by the richness of the related signals (here, the size of the regressor). The "signal to noise ratio" for this example can be thought as $SNR = \sqrt{\frac{\frac{1}{t}\sum_{i=1}^{i=t}(\varphi(i)\theta^0)^2}{\sigma^2}} = \frac{\sqrt{\frac{1}{t}\sum_{i=1}^{i=t}\varphi(i)^2}}{\sigma}\theta^0$ Obviously, the larger the SNR is, the smaller the co-variance matrix will be, and the better the resulting parameter estimate. So it is very important to look for practical ways to minimize SNR such as using as large inputs as possible to make $\varphi(t)$ bigger while at the same time making sure that the model structure assumptions are not violated to keep the model error $e(t)$ small.

When several parameters are estimated, the convergence rates may be different for different parameters, which are related to the structure of the matrix $\mathbf{P}(t) = (\mathbf{\Phi}(t)^T\mathbf{\Phi}(t))^{-1}$ and may be inferred based on the eigenvalues and the eigenvectors of $\mathbf{R}_{\boldsymbol{\varphi}}(0) = \sum_{j=1}^{p} \lambda_j v_j v_j^T$; the eigenvalue λ_j reveals the convergence rate while the eigenvector v_j describes the corresponding direction (i.e., $v_j^T\boldsymbol{\theta}$, a fixed linear combination of the model parameters) that converges at the rate. If $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ is near singular, then at least one of its eigenvalues is near zero, indicating that some linear combinations of the model parameters described by the corresponding eigenvector will not converge or convergence very slowly. ♢

Example 2. Consider the following system:

$$\begin{aligned} y(t) &= b_0 + b_1 u(t) + b_2 u(t)^2 + e(t), & e(t) &= \sigma \cdot \text{randn}, \\ &= \boldsymbol{\varphi}_a(t)^T\boldsymbol{\theta}_a + e(t), & \boldsymbol{\varphi}_a(t)^T &= \begin{bmatrix} 1 & u(t) & u^2(t) \end{bmatrix} \\ & & \boldsymbol{\theta}_a &= \begin{bmatrix} b_0 & b_1 & b_2 \end{bmatrix}^T \end{aligned} \tag{38}$$

where $e(t)$ is a Gaussian distributed zero mean white noise sequence with a standard deviation of σ. For this system, supposed that we do not know the above actual system structure and would like to use the following four sets of different model structures to fit the actual data respectively: All four model

	Assumed Model Structure	Regressor and Model Parameters
M1	$\hat{y}(t,\theta_I) = c_0$ $= \varphi_I(t)\theta_I$	$\varphi_I(t) = 1$ $\theta_I = c_0$
M2	$\hat{y}(t,\boldsymbol{\theta}_{II}) = c_0 + c_1 u(t)$ $= \boldsymbol{\varphi}_{II}(t)^T \boldsymbol{\theta}_{II}$	$\boldsymbol{\varphi}_{II}(t) = [1,\ u(t)]^T$ $\boldsymbol{\theta}_{II} = [c_0,\ c_1]^T$
M3	$\hat{y}(t,\boldsymbol{\theta}_{III}) = c_0 + c_1 u(t) + c_2 u(t)^2$ $= \boldsymbol{\varphi}_{III}(t)^T \boldsymbol{\theta}_{III}$	$\boldsymbol{\varphi}_{III}(t) = [1,\ u(t),\ u(t)^2]^T$ $\boldsymbol{\theta}_{III} = [c_0,\ c_1,\ c_2]^T$
M4	$\hat{y}(t,\boldsymbol{\theta}_{IV}) = c_0 + c_1 u(t) + c_2 u(t)^2 + c_3 u(t)^3$ $= \boldsymbol{\varphi}_{IV}(t)^T \boldsymbol{\theta}_{IV}$	$\boldsymbol{\varphi}_{IV}(t) = [1,\ u(t),\ u(t)^2,\ u(t)^3]^T$ $\boldsymbol{\theta}_{IV} = [c_0,\ c_1,\ c_2,\ c_3]^T$

Table 1. Model structures

structures are linear with respect to their model parameters. Thus, the LSE algorithm (29) can be used to obtain their model parameters:

$$\hat{\boldsymbol{\theta}}_M(t) = \mathbf{P}_M(t)\left[\sum_{i=1}^{t} \boldsymbol{\varphi}_M(i)\mathbf{y}(i)\right], \quad \mathbf{P}_M(t) = \left[\sum_{i=1}^{t} \boldsymbol{\varphi}_M(i)\boldsymbol{\varphi}_M(i)^T\right]^{-1} \tag{39}$$

where $M = I, II, III, IV$ for each model respectively. From (38), the model errors associated with the above model structures are

	Model Error	Uncorrelated with Regressor?
M1	$\epsilon_\theta(t) = b_1 u(t) + b_2 u(t)^2 + e(t)$ with $\theta_I^0 = b_0$	neither white nor uncorrelated with $\varphi_I(t)$ in general
M2	$\epsilon_\theta(t) = b_2 u(t)^2 + e(t)$ with $\boldsymbol{\theta}_{II}^0 = [b_0,\ b_1]^T$	neither white nor uncorrelated with $\varphi_{II}(t)$ in general.
M3	$\epsilon_\theta(t) = e(t)$ with $\boldsymbol{\theta}_{III}^0 = [b_0,\ b_1,\ b_2]^T$	white and uncorrelated with $\varphi_{III}(t)$
M4	$\epsilon_\theta(t) = e(t)$ with $\boldsymbol{\theta}_{IV}^0 = [b_0,\ b_1,\ b_2,\ 0]^T$	white and uncorrelated with $\varphi_{IV}(t)$

Table 2. Model structure errors

Thus, when the actual system (38) is absence of random noise $e(t)$, the LSE (39) is expected to exactly recover the true model parameters $\boldsymbol{\theta}_{III}^0$ and $\boldsymbol{\theta}_{IV}^0$ for the model structure M3 and M4 respectively, as long as experimental conditions are controlled so that $\mathbf{P}_{III}(t)$ and $\mathbf{P}_{IV}(t)$ are finite for the collected experimental data set. This is confirmed by the LSE results shown in table 2 for $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $\sigma = 0$ in (38) when the actual output

is generated using an input sequence of $u(i) = i - 1$ for $i = 1, \ldots, t = 7$, an input sequence rich enough that $\mathbf{P}_M(t)$ exists for all four model structures.

Model	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))$
M1	2.6				14.56
M2	0.5	0.7			0.84
M3	1.0	0.1	0.1		0
M4	1.0	0.1	0.1	0	0

Table 3. LSE results in absence of random noises ($\sigma = 0$)

The above results are also shown in Fig. 3 in terms of fitting curves having the assumed linear regression form (the solid lines in the figures) to the measured data set (the dots in the figures):

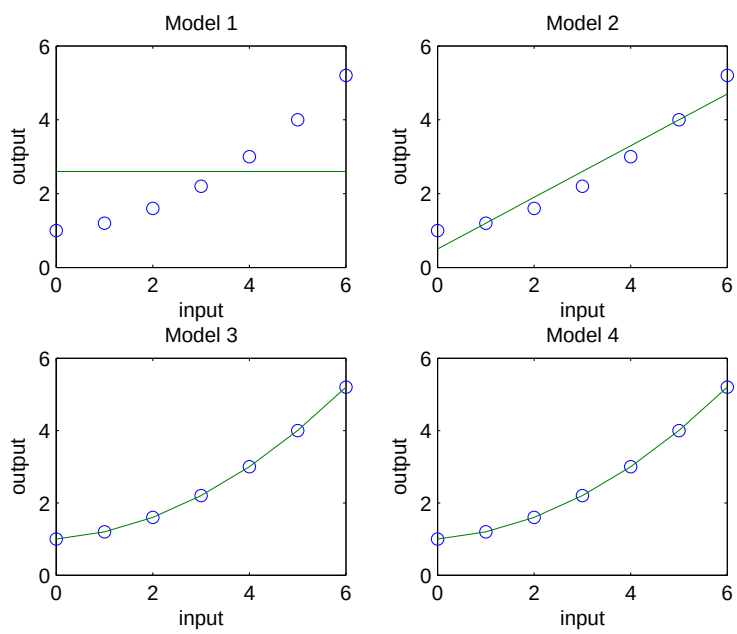

Fig. 3. Geometric interpretation of LSE in terms of curve fitting

When the actual system does have the random zero mean white noise $e(t)$, the LSE results with model structure M3 and M4 for four different runs with $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $\sigma = 0.1$ in (38) are shown in the following tables when the actual output is generated using an input sequence of $u(i) = i - 1$ for $i = 1, \ldots, t = 7$.

Table 4. LSE results for M3 with random noises ($\sigma = 0.1$) and $t = 7$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	1.01	0.11	0.097		0.0025
2	0.902	0.236	0.077		0.0081
3	0.900	0.169	0.094		0.0021
4	1.102	0.017	0.111		0.0054

Table 5. LSE results for M4 with random noises ($\sigma = 0.1$) and $t = 7$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	0.969	0.252	0.034	0.007	0.001
2	0.902	0.235	0.078	-0.0001	0.0081
3	0.877	0.245	0.060	0.004	0.0017
4	1.080	0.091	0.077	0.004	0.0050

As seen from the above table, as the model errors for both model structures are zero mean and uncorrelated with the regressor, the LSE for both model structures are unbiased – the LSE over different runs are on average the same as the true model parameter value of $b_0 = 1$, $b_1 = 0.1$, $b_2 = 0.1$, and $b_3 = 0$. In fact, for large t, as shown in tables 6 and 7 for $t = 1000$, the estimates are almost the same as their true values for each run as predicted by the theory. Furthermore, as the model structure M4 is larger than M3, a smaller fitting error index J always results in each run. However, this does not mean that the resulting parameter estimates are more accurate by comparing the parameter estimates of M3 and M4.

Table 6. LSE results for M3 with random noises ($\sigma = 0.1$) and $t = 1000$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	0.9955	0.1	0.1		0.0089
2	0.988	0.100	0.100		0.0106
3	1.010	0.100	0.100		0.0105
4	1.0007	0.1000	0.1000		0.0098

Table 7. LSE results for M4 with random noises ($\sigma = 0.1$) and $t = 1000$

Run	$\hat{b}_0(t)$	$\hat{b}_1(t)$	$\hat{b}_2(t)$	$\hat{b}_3(t)$	$J(t, \hat{\boldsymbol{\theta}}(t))/t$
1	1.005	0.100	0.100	0.000	0.0089
2	0.987	0.100	0.100	0.000	0.0106
3	1.012	0.100	0.100	0.000	0.0105
4	0.9939	0.1001	0.1000	0.0000	0.0098

This example shows that it is important to choose the correct model structure in order to get a good fit of data and a consistent and accurate estimate of true model parameters. With a too small model structure or few model parameters to tune such as the model structures M1 and M2 in this example, it

is not possible to get a good fit of the data, left alone the accurate estimate of true model parameters. On the other hand, with a too large model structure or too many model parameters to be determined (i.e., p is large), more complicated experimental conditions are needed to perform the LSE algorithm and to guarantee the convergence of the LSE estimates. Furthermore, though the fit to the measured data would be better (e.g., the model structure M4), the resulting parameter estimates do not necessarily be better and the fit to another data set may be worse (*overfitting* or fit to the noise). ♢

2.5 Recursive Least Squares Estimation (RLSE)

Implementation of LSE (29) requires a large number of matrix multiplications and summations and, worst of all, the inversion of a $p \times p$ matrix in (30) at each time instance t, which could be very time-consuming when the number of parameters to be estimated in practice is normally large. To by-pass this computation problem, we will use the following Matrix Inversion Lemma (MIL) which converts the inversion computation of a matrix of larger dimension (i.e., $A + BCD$) to the inversion problem of a smaller dimension matrix (i.e., $C^{-1} + DA^{-1}B$):

Lemma 2. Matrix Inversion Lemma (MIL)
Let $\mathbf{A}$, $\mathbf{C}$, *and* $(\mathbf{C}^{-1} + \mathbf{D}\mathbf{A}^{-1}\mathbf{B})$ *be nonsingular square matrices. Then* $\mathbf{A} + \mathbf{BCD}$ *is invertible, and*

$$(\mathbf{A} + \mathbf{BCD})^{-1} = \mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{D}\mathbf{A}^{-1}\mathbf{B})^{-1}\mathbf{D}\mathbf{A}^{-1} \tag{40}$$

△

Assuming that the weighting matrices in (24) are chosen as

$$\mathbf{W}(t,t) = I_m, \quad \mathbf{W}(t,i) = \prod_{k=i+1}^{t} \lambda(k) I_m, \ \forall t, i \tag{41}$$

where $\lambda(k)$ is any positive number representing the forgetting factor, then, from (30)

$$\begin{aligned}
\mathbf{P}(t)^{-1} &= \sum_{i=1}^{t} \boldsymbol{\varphi}(i)\mathbf{W}(t,i)\boldsymbol{\varphi}(i)^T = \sum_{i=1}^{t-1} \left(\prod_{k=i+1}^{t} \lambda(k) \right) \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T \\
&= \lambda(t) \sum_{i=1}^{t-1} \left(\prod_{k=i+1}^{t-1} \lambda(k) \right) \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T \\
&= \lambda(t) \sum_{i=1}^{t-1} \boldsymbol{\varphi}(i)\mathbf{W}(t-1,i)\boldsymbol{\varphi}(i)^T + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T \\
&= \lambda(t)\mathbf{P}(t-1)^{-1} + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T
\end{aligned} \tag{42}$$

Applying MIL with $\mathbf{A} = \lambda(t)\mathbf{P}(t-1)^{-1}$, $\mathbf{B} = \boldsymbol{\varphi}(t)$, $\mathbf{C} = \mathbf{I}_m$, and $\mathbf{D} = \boldsymbol{\varphi}(t)^T$, $\mathbf{P}(t)$ can be recursively calculated by

$$\mathbf{P}(t) = \tfrac{1}{\lambda(t)}\mathbf{P}(t-1)\left\{I - \boldsymbol{\varphi}(t)\left[I_m + \boldsymbol{\varphi}(t)^T \tfrac{1}{\lambda(t)}\mathbf{P}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\boldsymbol{\varphi}(t)^T \tfrac{1}{\lambda(t)}\mathbf{P}(t-1)\right\} \tag{43}$$

Noting that the LSE (29) can be rewritten as

$$\mathbf{P}(t)^{-1}\hat{\boldsymbol{\theta}}(t) = \sum_{i=1}^{t} \boldsymbol{\varphi}(i)\mathbf{W}(t,i)\mathbf{y}(i) \tag{44}$$

and (42), with the weighting given by (41), it is straightforward to verify that

$$\begin{aligned}\sum_{i=1}^{t} \boldsymbol{\varphi}(i)\mathbf{W}(t,i)\mathbf{y}(i) &= \sum_{i=1}^{t-1} \boldsymbol{\varphi}(i)\lambda(t)\mathbf{W}(t-1,i)\mathbf{y}(i) + \boldsymbol{\varphi}(t)\mathbf{y}(t) \\ &= \lambda(t)\sum_{i=1}^{t-1} \boldsymbol{\varphi}(i)\mathbf{W}(t-1,i)\mathbf{y}(i) + \boldsymbol{\varphi}(t)\mathbf{y}(t) \\ &= \lambda(t)\mathbf{P}(t-1)^{-1}\hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{\varphi}(t)\mathbf{y}(t) \\ &= \mathbf{P}(t)^{-1}\hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{\varphi}(t)[\mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1)]\end{aligned} \tag{45}$$

Substituting (45) into (29), the following **recursive formula for LSE** is obtained:

$$\boxed{\begin{aligned}&\boldsymbol{\Gamma}(t-1) = \tfrac{1}{\lambda(t)}\mathbf{P}(t-1) \\ &\mathbf{P}(t) = \boldsymbol{\Gamma}(t-1)\left\{I - \boldsymbol{\varphi}(t)\left[I_m + \boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\right\} \\ &\hat{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t-1) + \mathbf{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\epsilon}^0(t), \qquad \boldsymbol{\epsilon}^0(t) := \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1)\end{aligned}} \tag{46}$$

in which the intermediate adaptation rate matrix $\boldsymbol{\Gamma}(t-1)$ is introduced for notational simplicity and efficient computation, and $\boldsymbol{\epsilon}^0(t)$ is normally called the *a priori* prediction error [5], i.e., the prediction error using previous parameter estimate $\hat{\boldsymbol{\theta}}(t-1)$.

The above recursive formula only involves the inversion of a $m \times m$ matrix, a much smaller dimension matrix inversion operation than the original LSE algorithm due to the fact that the number of outputs m is normally much smaller than the number of parameters p to be estimated. In fact, for a single output system, the inversion simply reduces to a division. (46) also shows that the covariance matrix $\mathbf{P}(t)$ can be thought as the adaptation gain matrix at each sampling instance.

The initial conditions to start the above recursive formula can be obtained in the following two ways:

1. Pick a t_0 large enough such that $\boldsymbol{\Phi}^T(t_0)\boldsymbol{\Phi}(t_0)$ is nonsingular. Calculate $\mathbf{P}(t_0)$ and $\hat{\boldsymbol{\theta}}(t_0)$ using the LSE algorithm (29) and (30), and start the recursive formula (46) to calculate $\hat{\boldsymbol{\theta}}(t)$ for $t \geq t_0$. This will lead to exactly the same parameter estimates as the original LSE algorithm. However, this initialization needs to wait for some time before an estimate can be calculated.
2. As an alternative, an arbitrarily large symmetric positive definite (s.p.d.) matrix $\mathbf{P}_0 > 0$ can be chosen. Let $\mathbf{P}(0) = \mathbf{P}_0$ and $\hat{\boldsymbol{\theta}}(0) = \hat{\boldsymbol{\theta}}_0$ to start the

RLS algorithm, where $\hat{\boldsymbol{\theta}}_0$ is any initial parameter estimate. In this case, the RLS estimate is slightly different from the original LSE estimate; it actually calculates $\boldsymbol{\theta}$ that minimizes the following performance index

$$J(t,\boldsymbol{\theta}) = \sum_{i=1}^{t} \boldsymbol{\epsilon}_\theta(i)^T \mathbf{W}(t,i)\boldsymbol{\epsilon}_\theta(i) + \left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_0\right)^T \mathbf{P}_0^{-1}\left(\boldsymbol{\theta} - \hat{\boldsymbol{\theta}}_0\right) \tag{47}$$

With $P_0^{-1} \approx 0$ or $P_0 \gg 1$, the performance index (47) is almost the same as the LSE performance index (24), which is the rational behind this initialization procedure. However, in the absence of model error, unlike the original LSE, parameter estimates with this initialization will not be exactly the same as the actual values even when $\boldsymbol{\Phi}^T(t_0)\bar{\mathbf{W}}(t_0)\boldsymbol{\Phi}(t_0)$ is nonsingular for some $t_0 \le t$. In fact, this initialization with the recursive formula (46) leads to $\tilde{\boldsymbol{\theta}}(t) = \mathbf{P}(t)\mathbf{P}(0)^{-1}\tilde{\boldsymbol{\theta}}(0)$, which will not be zero unless $\mathbf{P}(t) \to 0$ as $t \to \infty$, a condition that can be met only when the PE condition is satisfied.

For analysis, it is normally useful to rewrite the above recursive formula in terms of the so called *a posteriori* prediction error $\boldsymbol{\epsilon}(t)$ [5] defined by

$$\boldsymbol{\epsilon}(t) := \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t) \tag{48}$$

i.e., the prediction error using current parameter estimate $\hat{\boldsymbol{\theta}}(t)$. Substituting (46) into (48), $\boldsymbol{\epsilon}(t)$ is related to $\boldsymbol{\epsilon}^0(t)$ by

$$\begin{aligned}\boldsymbol{\epsilon}(t) &= \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1) - \boldsymbol{\varphi}(t)^T\mathbf{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\epsilon}^0(t)\\ &= \left[I_m - \boldsymbol{\varphi}(t)^T\mathbf{P}(t)\boldsymbol{\varphi}(t)\right]\boldsymbol{\epsilon}^0(t)\end{aligned} \tag{49}$$

Noting the recursive formula for $\mathbf{P}(t)$ in (46), it is straightforward to verify that the relationship between $\boldsymbol{\epsilon}(t)$ and $\boldsymbol{\epsilon}^0(t)$ can also be written as

$$\boldsymbol{\epsilon}(t) = \left[I_m + \boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\boldsymbol{\epsilon}^0(t) \tag{50}$$

From (46),

$$\begin{aligned}\mathbf{P}(t)\boldsymbol{\varphi}(t) &= \mathbf{P}(t-1)\boldsymbol{\varphi}(t)\left[\lambda(t)I_m + \boldsymbol{\varphi}(t)^T\mathbf{P}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\\ &= \boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\left[I_m + \boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\end{aligned} \tag{51}$$

Substituting (50) and (51) into (46), the parameter update law can also be written in terms of $\boldsymbol{\epsilon}(t)$ as

$$\begin{aligned}&\hat{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\epsilon}(t)\\ &\boldsymbol{\epsilon}(t) = \left[I_m + \boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\right]^{-1}\boldsymbol{\epsilon}^0(t),\ \boldsymbol{\epsilon}^0(t) = \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1)\end{aligned} \tag{52}$$

2.6 On-line Parameter Estimation Algorithms

On-line estimation of parameters of a plant model or a controller is one of the key steps in building an adaptive control system. This section focuses on the typical issues associated with on-line parameter estimation and presents some common strategies and specific estimation algorithms.

The two main issues for on-line parameter estimation are: (i) Parameter estimation algorithms need to be computationally efficient for real-time implementation; and (ii) Experimental conditions may not be controlled as well as in off-line estimation and the resulting data set may not be informative and rich enough, possibly even during some large intervals of time.

The RLSE algorithm (46) solves the computation problem. Its suitability as an on-line parameter estimator will thus be carefully examined.

RLSE without Forgetting Factor

The first case is the RLS algorithm (46) without any weighting (or forgetting factor), i.e., let $\lambda(t) = 1$ in (41). In this case, $\forall t$,

$$\begin{aligned} & \mathbf{P}^{-1}(t) = \mathbf{P}^{-1}(t-1) + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T \geq \mathbf{P}^{-1}(t-1) \\ \Rightarrow \quad & \mathbf{P}(t-1) \geq \mathbf{P}(t) \ \rightarrow \ \mathbf{P}_\infty \geq 0 \quad \text{as} \quad t \rightarrow \infty \end{aligned} \tag{53}$$

Thus, the RLS algorithm can always be run without any computational issues as $0 \leq \mathbf{P}(t) \leq \mathbf{P}(0), \forall t$. Furthermore, if the following *persistent excitation* (PE) condition is satisfied,

$$\exists t_0, T, \alpha > 0 \ \text{ such that (s.t.)} \quad \sum_{i=t+1}^{t+T} \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T \geq \alpha I_p, \ \forall t \geq t_0 \tag{54}$$

then, $\forall k$, $\mathbf{P}^{-1}(t_0 + kT) \geq \mathbf{P}^{-1}(t_0) + k\alpha I_p \geq k\alpha I_p$ and $\mathbf{P}(t_0 + kT) \leq \frac{1}{k\alpha} I_p$. So, $\mathbf{P}(t) \rightarrow 0$ as $t \rightarrow \infty$, which implies that parameter estimation (46) will eventually cease to update. Thus, this RLS algorithm is only good for on-line estimation of parameters that will never change over time.

RLSE with Forgetting Factor and Covariance Limiting

A main objective of on-line parameter estimation is to track slow changes of the time-varying parameters. To achieve this goal, it is necessary to discard old data and place more emphasis on new data. One simple way to achieve this is to choose $\lambda(t)$ in (41) as any positive number less than 1. For simplicity, let $\lambda(t)$ be a constant that is less than 1, $\lambda < 1$. Then, $\mathbf{W}(t,k) = \lambda^{t-k} I_m$, which exponentially converges to zero as $t - k \rightarrow \infty$, achieving the goal of discarding old data (large $t - k$). The smaller the forgetting factor λ, the faster the old information will be forgotten, which can lead to on-line parameter estimation that adapts quickly when the actual parameters change. However, such an algorithm also suffers from the following practical problems:

1. The covariance matrix $\mathbf{P}(t)$ may grow (wind-up) exponentially at a rate of $1/\lambda$ when the signal excitation is poor. For example, with no excitation, i.e., $\boldsymbol{\varphi}(t) = 0$, from (46), $\mathbf{P}(t) = (1/\lambda)^t \mathbf{P}(0) \to \infty$ as $t \to \infty$. This is in contrast with the RLS without forgetting factor where the covariance matrix always decreases with time.
2. The parameter estimates are more sensitive to model-error or noise since essentially less number of data are used to smooth out their noise effects due to the discard of old data.

In practice, the selection of an appropriate forgetting factor can be facilitated by the equivalent time constant concept:

$$e^{-\frac{h}{T_f}} = \lambda \quad \Rightarrow \quad T_f = -\frac{h}{\ln \lambda} \approx \frac{h}{1-\lambda} = \begin{cases} 100h \text{ for } \lambda = 0.99 \\ 10h \qquad \lambda = 0.9 \end{cases} \tag{55}$$

where h is the sampling period and T_f is the equivalent time constant. It is thus clear that with a forgetting factor of $\lambda = 0.99$, approximately 100 to 400 immediately previous data sets are used by the current parameter estimation. If the actual parameters do not change much within 400 sample periods, such a forgetting factor might be appropriate. Moreover, a forgetting factor of $\lambda = 0.9$ leads to an RLS algorithm that essentially only uses the past 10 to 40 sample data in the parameter estimates. These estimates could be very noise (or model error) sensitive and have a greater risk of incurring the covariance matrix wind-up problem. Thus, when choosing the forgetting factor, a trade-off has to be made between the ability to track time-varying parameters and the parameter estimation noise/model error sensitivity.

To by-pass the RLS algorithm with forgetting factor's covariance wind-up problem, the covariance limiting should be used in practice; the parameter estimate update (46) does not change but the covariance matrix update is changed to

$$\boxed{\begin{aligned} \bar{\mathbf{P}}(t) &= \boldsymbol{\Gamma}(t-1) - \boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}\left[I_m + \boldsymbol{\varphi}^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}\right]^{-1}\boldsymbol{\varphi}^T\boldsymbol{\Gamma}(t-1) \\ \mathbf{P}(t) &= \begin{cases} \mathbf{P}(t-1) & \text{if} \quad \lambda_M\left(\bar{\mathbf{P}}(t)\right) > \rho_M \\ \bar{\mathbf{P}}(t) & \text{otherwise (i.e., } \bar{\mathbf{P}}(t) \le \rho_M I_p \text{)} \end{cases} \end{aligned}} \tag{56}$$

where $\boldsymbol{\Gamma}(t-1) = \frac{1}{\lambda(t)}\mathbf{P}(t-1)$ with $\lambda(t)$ being the forgetting factor, $\lambda_M(\bullet)$ represents the maximum eigenvalue of a symmetric matrix $\bullet$, and ρ_M are the pre-set upper limit for $P(t)$. With this modification, $P(t) \le \rho_M I_p, \ \forall t$.

RLSE with Covariance Re-setting

Another simple way to discard old data is to use covariance re-setting [3]. This method may also be used when parameters are constant over long periods of time and then suddenly change (e.g., grasping or dumping a payload); the covariance matrix can be reset to a large positive definite matrix if the change can be detected (e.g., analyzing residuals or rate of parameter estimation

changes). Covariance resetting essentially discards all past data and enables RLS to track the sudden change quickly, otherwise it takes at least around the order of the forgetting factor equivalent time constant for a RLS estimate to converge to a new value.

Let $Z_r = \{t_{r1}, t_{r2}, \ldots\}$ be the set of times at which the covariance matrix needs to be re-set; t_{ri} can be a fixed time instance pre-specified by the user or a varying time instance that occurs when the on-line covariance matrix update drops below a pre-set threshold. Then, the covariance matrix is updated like in RLS without forgetting factor, i.e., (53), except at time instance t_{ri} the co-variance matrix is re-set to some pre-specified diagonal matrix $\rho_{ri} I_p > 0$.

General Parameter Adaptation Algorithms (PAA)

The following PAA is proposed, which combines covariance limiting and the MIMO generalization of the general structure for on-line parameter adaptation algorithms (PAA) presented in [5],

$$\begin{aligned}
&\hat{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t-1) + \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t), \quad \boldsymbol{\nu}(t) = \left[I_m + \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\right]^{-1} \boldsymbol{\nu}^0(t) \\
&\bar{\boldsymbol{\Gamma}}(t)^{-1} = \lambda_1(t)\boldsymbol{\Gamma}(t-1)^{-1} + \lambda_2(t)\boldsymbol{\phi}(t)\boldsymbol{\phi}(t)^T, \quad 0 < \Gamma(0) < \rho_M I_p \\
&\boldsymbol{\Gamma}(t) = \begin{cases} \boldsymbol{\Gamma}(t-1), & \text{if} \quad \lambda_M\left(\bar{\boldsymbol{\Gamma}}(t)\right) > \rho_M \\ \bar{\boldsymbol{\Gamma}}(t) & \text{otherwise} \end{cases}
\end{aligned} \tag{57}$$

in which $\boldsymbol{\Gamma}(t-1)$ is the adaptation gain matrix, $\boldsymbol{\phi}(t)$ is an observation vector similar to the regressor vector $\boldsymbol{\varphi}(t)$ in (46), $\boldsymbol{\nu}(t)$ is the *a posteriori* adaptation error, $\boldsymbol{\nu}^0(t)$ is the *a priori* adaptation error, and $0 < \lambda_1(t) \leq 1$ and $0 \leq \lambda_2(t) < 2$ for stability of the resulting algorithm. The *a priori* adaptation error $\nu^0(t)$ depends only on the parameter estimates $\hat{\boldsymbol{\theta}}(i)$ up to and including $i = t-1$ for the algorithm to be implementable. Furthermore, it is chosen so that the *a posteriori* adaptation error vector $\boldsymbol{\nu}(t)$ calculated from (57) are related to the prediction error vector by:

$$\boldsymbol{\nu}(t) = \mathbf{H}\left(q^{-1}\right)[\boldsymbol{\epsilon}(t)], \quad \boldsymbol{\epsilon}(t) = \mathbf{y}(t) - \boldsymbol{\phi}(t)^T \hat{\boldsymbol{\theta}}(t) \tag{58}$$

where $\mathbf{H}\left(q^{-1}\right)$ is a matrix of rational discrete transfer operators. In addition, for stability, it is required that the resulting transfer function matrix

$$\mathbf{H}'\left(z^{-1}\right) = \mathbf{H}\left(z^{-1}\right) - \tfrac{\lambda_H}{2} I_m \tag{59}$$

be strictly positive real (SPR) for some λ_H satisfying $\lambda_H > \lambda_{2M} = \sup_t \{\lambda_2(t)\}$. Relaxing this SPR requirement in SISO case was carried out by Tomizuka in [8, 9].

Theorem 1. *Assume that the* a posteriori *adaptation error vector satisfies (58) where $\boldsymbol{\phi}(t)$ is a bounded or unbounded matrix sequence and the transfer function (59) is strictly positive real. Then, when the actual output can be represented as*

$$\mathbf{y}(t) = \boldsymbol{\phi}(t)^T \boldsymbol{\theta} \tag{60}$$

for a constant $\boldsymbol{\theta}$*, the PAA (57) guarantees that*

P1 . The parameter estimation error $\tilde{\boldsymbol{\theta}}(t) := \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}$ *is uniformly bounded. Furthermore, when* $H\left(q^{-1}\right) = I_m$*,*

$$V_\theta(t) = \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t)^{-1} \tilde{\boldsymbol{\theta}}(t) \leq V_\theta(0) = \tilde{\boldsymbol{\theta}}(0)^T \boldsymbol{\Gamma}(0)^{-1} \tilde{\boldsymbol{\theta}}(0) \tag{61}$$

P2 . $\boldsymbol{\epsilon}, \boldsymbol{\nu} \in l_2^m$*, i.e.,* $\lim_{t\to\infty} \sum_{i=1}^{t} \|\boldsymbol{\epsilon}(i)\|^2$ *and* $\lim_{t\to\infty} \sum_{i=1}^{t} \|\boldsymbol{\nu}(i)\|^2$ *exist, and*

$$\begin{aligned} &\lim_{t\to\infty} \sum_{i=1}^{t} \boldsymbol{\nu}(i)^T \left[I_m + \boldsymbol{\phi}(i)^T \boldsymbol{\Gamma}(i-1)\boldsymbol{\phi}(i)\right] \boldsymbol{\nu}(i) \\ &\quad = \lim_{t\to\infty} \sum_{i=1}^{t} \boldsymbol{\nu}^0(i)^T \left[I_m + \boldsymbol{\phi}(i)^T \boldsymbol{\Gamma}(i-1)\boldsymbol{\phi}(i)\right]^{-1} \boldsymbol{\nu}^0(i) < \infty \end{aligned} \tag{62}$$

Thus, the a posteriori *adaptation and prediction errors are bounded and go to zero asymptotically:*

$$\lim_{t\to\infty} \boldsymbol{\nu}(t) = \lim_{t\to\infty} \boldsymbol{\epsilon}(t) = \lim_{t\to\infty} \sqrt{I_m + \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)}\boldsymbol{\nu}(t) = 0 \tag{63}$$

P3 . $\{\hat{\boldsymbol{\theta}}(t) - \hat{\boldsymbol{\theta}}(t-1)\} \in l_2^p$*, and for any finite* k*,*

$$\lim_{t\to\infty} \|\hat{\boldsymbol{\theta}}(t) - \hat{\boldsymbol{\theta}}(t-k)\| = 0 \tag{64}$$

△

The theorem gives the stability and asymptotic convergence of the adaptation and prediction errors without model errors. However, the theorem does not guarantee the asymptotic convergence of parameter estimation error $\tilde{\boldsymbol{\theta}}(t)$; not even the convergence of the parameter estimate $\hat{\boldsymbol{\theta}}(t)$ to a constant value. For asymptotic convergence of $\tilde{\boldsymbol{\theta}}(t)$ to zero, additional conditions such as the *persistent exciting* condition introduced later are needed.

Using the MIL, the adaptation gain matrix in (57) can be implemented as

$$\bar{\boldsymbol{\Gamma}}(t) = \tfrac{1}{\lambda_1(t)}\boldsymbol{\Gamma}(t-1)\left\{I_p - \boldsymbol{\phi}(t)\left[\tfrac{\lambda_1(t)}{\lambda_2(t)} I_m + \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\right]^{-1} \boldsymbol{\phi}^T(t)\boldsymbol{\Gamma}(t-1)\right\} \tag{65}$$

When $\boldsymbol{\nu}^0(t)$ is chosen as the *a priori* prediction error $\boldsymbol{\epsilon}^0(t)$ in (46), with $\boldsymbol{\phi}(t) = \boldsymbol{\varphi}(t)$, it is easy to show that

$$\boldsymbol{\nu}(t) = \boldsymbol{\epsilon}(t) = \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T \hat{\boldsymbol{\theta}}(t) \tag{66}$$

Thus, (58) is satisfied with $H = 1$ and (59) is true for any $\lambda_H < 2$. In this sense, (57) can also be thought as the RLS algorithm generalized by introducing two weighting sequences $\lambda_1(t)$ and $\lambda_2(t)$ into the covariance matrix updating law. Some choices of the sequences $\lambda_1(t)$ and $\lambda_2(t)$ and their interpretations are

A1 . *Vanishing gain (RLS without forgetting factor).* By choosing

$$\lambda_1(t) = \lambda_1 = 1; \quad \lambda_2(t) = \lambda_2 = 1 \tag{67}$$

the general PAA algorithm becomes the RLS without forgetting factor. The covariance matrix (or adaptation gain matrix) exponentially converges to zero when the PE condition is satisfied.

A2 . *RLS with constant forgetting factor.* By choosing

$$\lambda_1(t) = \lambda_2(t) = \lambda_0 < 1 \tag{68}$$

it is easy to verify that the general PAA algorithm becomes the RLS with a forgetting factor of λ_0 with a co-variance matrix of $\mathbf{P}(t) = \lambda_0 \mathbf{\Gamma}(t)$.

A3 . *RLS with variable forgetting factor.* By choosing

$$\lambda_1(t) = \lambda_2(t) = \lambda_{LS}(t+1) < 1 \tag{69}$$

it can be verified that the general PAA algorithm becomes the RLS with a time-varying forgetting factor of $\lambda_{LS}(t)$ and a co-variance matrix of $\mathbf{P}(t) = \lambda_{LS}(t+1)\mathbf{\Gamma}(t)$. Thus, by choosing

$$\lambda_{LS}(t) = \lambda_0 \lambda_{LS}(t-1) + 1 - \lambda_0, \quad 0 < \lambda_0 < 1 \tag{70}$$

with typical values of $\lambda_{LS}(0) = 0.95$ to 0.99 and $\lambda_0 = 0.5$ to 0.99, a varying forgetting factor that starts with an initial faster forgetting of a $\lambda_{LS}(0)$ much less than 1 and asymptotically tends towards 1 for no-forgetting as $t \to \infty$ is obtained. Because of this, only the initial data are forgotten and the covariance matrix (or adaptation gain) tends towards a decreasing gain. Overall, it shares the same property as the RLS without forgetting factor, but avoids a too rapid decrease of the adaptation gain since the initial forgetting factor is less than 1. This generally accelerates the convergence by maintaining high gain at the beginning when the estimates are far from their optimal values. Thus, this type of profile is recommended for the model identification of stationary systems.

Other evolutions of $\lambda_1(t)$ can also be considered. For example,

$$\lambda_{LS}(t) = 1 - \alpha \frac{\left[\nu^0(t)\right]^2}{1+\boldsymbol{\phi}(t)^T \mathbf{P}(t-1)\boldsymbol{\phi}(t)}; \quad \alpha > 0 \tag{71}$$

The forgetting factor tends towards 1 when the prediction error tends towards zero. However, when a change occurs in the system parameters, the prediction error increase leads to a forgetting factor less than 1 to assure good adaptation capability.

A4 . *Constant gain or gradient algorithm.* By choosing

$$\lambda_1(t) = \lambda_1 = 1; \quad \lambda_2(t) = \lambda_2 = 0 \tag{72}$$

from (57), $P(t) = P(t-1) = P(0)$. Thus a constant adaptation gain matrix is implemented, normally referred to as the gradient algorithm. This

algorithm is simpler to implement but normally results in performance inferior to other RLS type algorithms. As such, it is normally only used for the identification and adaptive control of stationary or time-varying systems with few parameters and a reduced noise level.

2.7 Parameter Estimation in Continuous Time Domain

So far we have thoroughly covered the parameter estimation for linear regression type prediction models in discrete time domain. In this subsection, the corresponding results in continuous time domain will be detailed.

Consider the following linear regression model in continuous time domain

$$\hat{\mathbf{y}}(t,\boldsymbol{\theta}) = \boldsymbol{\varphi}_1(t)\theta_1 + \boldsymbol{\varphi}_2(t)\theta_2 + \cdots + \boldsymbol{\varphi}_p(t)\theta_p = \boldsymbol{\varphi}(t)^T\boldsymbol{\theta} \tag{73}$$

where $\hat{\mathbf{y}} \in R^m$. The goal is to determine model parameters $\boldsymbol{\theta}$ in (73) so that the predicted output $\hat{\mathbf{y}}$ overall fits the measured output $\mathbf{y}$ well as measured by minimizing the following weighted least-squares type performance index function

$$J(t,\boldsymbol{\theta}) = \int_0^t \left\|\mathbf{R}(t,\tau)\left[\mathbf{y}(\tau) - \boldsymbol{\varphi}(\tau)^T\boldsymbol{\theta}\right]\right\|^2 d\tau = \int_0^t \boldsymbol{\epsilon}_\theta(\tau)^T\mathbf{W}(t,\tau)\boldsymbol{\epsilon}_\theta(\tau)d\tau \tag{74}$$

where $\mathbf{R}(t,\tau)$ or $\mathbf{W}(t,\tau)$ represents the weighting matrix, $\boldsymbol{\epsilon}_\theta(\tau) = \mathbf{y}(\tau) - \hat{\mathbf{y}}(\tau,\boldsymbol{\theta})$ is the model prediction error at time τ with model parameter vector $\boldsymbol{\theta}$. The necessary condition for optimality, i.e., the corresponding normal equation (75) in continuous time domain becomes

$$\left[\underbrace{\int_0^t \boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\boldsymbol{\varphi}(\tau)^T d\tau}_{\mathbf{R}(t)}\right]\hat{\boldsymbol{\theta}}(t) = \int_0^t \boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\mathbf{y}(\tau)d\tau \tag{75}$$

Thus, when $\mathbf{R}(t)$ is nonsingular, the minimization solution is unique and the resulting LSE can be calculated by

$$\hat{\boldsymbol{\theta}}(t) = \mathbf{P}(t)\int_0^t \boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\mathbf{y}(\tau)d\tau \tag{76}$$

where $\mathbf{P}(t) = \mathbf{R}(t)^{-1}$ is the *covariance matrix*.

Theorem 2. *Consider the following weighting*

$$\mathbf{W}(t,\tau) = e^{-\int_\tau^t \mu(s)ds} I_m \tag{77}$$

where $\mu(s) \geq 0$ is any non-negative function. Then the LSE (76) can be calculated by the following recursive formula

$$\begin{aligned} &\dot{\hat{\boldsymbol{\theta}}}(t) = \mathbf{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\epsilon}(t), \quad \boldsymbol{\epsilon}(t) := \mathbf{y}(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t) \\ &\dot{\mathbf{P}}(t) = \mu(t)\mathbf{P}(t) - \mathbf{P}(t)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T\mathbf{P}(t) \end{aligned} \tag{78}$$

△

Remark 1. With $\mu(t) = \alpha, \forall t$, the weighting (77) becomes $\mathbf{W}(t,\tau) = e^{-\alpha(t-\tau)}$, which corresponds to the usual exponential forgetting discussed in [2]. $\diamondsuit$

Remark 2. In implementation, since certain discretization has to be used, direct updating $\mathbf{P}(t)$ by (78) may have some numerical discretization problems as it contains quadratic term of $\mathbf{P}(t)$ which might be sensitive to discretization error when it is large. As an alternative, as shown in the proof for the theorem, we can update $\mathbf{P}(t)$ indirectly by calculating its inverse $\mathbf{R}(t) = \mathbf{P}^{-1}(t)$ as

$$\dot{\mathbf{R}}(t) = -\mu(t)\mathbf{R}(t) + \boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T \tag{79}$$

which is linear with respect to $\mathbf{R}(t)$ and may be less sensitive to discretization errors. $\diamondsuit$

Similar to (57), the above continuous RLSE algorithm can be generalized by introducing two non-negative functions $\mu_1(t) \geq 0$ and $\mu_2(t)$ and using covariance limiting in updating the covariance matrix as

$$\begin{aligned}
&\dot{\hat{\boldsymbol{\theta}}}(t) = \boldsymbol{\Gamma}(t)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t), \\
&\mathbf{D}_\Gamma = \mu_1(t)\boldsymbol{\Gamma}(t) - \mu_2(t)\boldsymbol{\Gamma}(t)\boldsymbol{\phi}(t)\boldsymbol{\phi}(t)^T\boldsymbol{\Gamma}(t) \\
&\dot{\boldsymbol{\Gamma}}(t) = \begin{cases} \mathbf{D}_\Gamma, & \text{if } \lambda_M(\boldsymbol{\Gamma}(t)) < \rho_M \ \text{ or } \ \mathbf{v}_M^T\mathbf{D}_\Gamma\mathbf{v}_M < 0 \\ 0 & \text{else} \end{cases}
\end{aligned} \tag{80}$$

where $\boldsymbol{\phi}(t)$ is an observation vector, and $\boldsymbol{\nu}(t)$ is the adaptation error vector which is assumed to related to the prediction error vector by

$$\boldsymbol{\nu}(t) = \mathbf{H}(p)\left[\boldsymbol{\epsilon}(t)\right], \quad \boldsymbol{\epsilon}(t) = \mathbf{y}(t) - \boldsymbol{\phi}(t)^T\hat{\boldsymbol{\theta}}(t) \tag{81}$$

in which $\mathbf{H}(p)$ is a matrix of rational transfer operators. In addition, for stability, it is required that $0 \leq \mu_2(t) < 2$ and the resulting transfer function matrix

$$\mathbf{H}'(s) = \mathbf{H}(s) - \tfrac{\mu_H}{2} I_m \tag{82}$$

be strictly positive real (SPR) for some μ_H satisfying $\mu_H > \mu_{2M} = \sup_t \{\mu_2(t)\}$. In (80), $\mathbf{v}_M$ represents the eigenvector corresponding to $\lambda_M(\boldsymbol{\Gamma}(t))$ when $\lambda_M(\boldsymbol{\Gamma}(t)) = \rho_M$, and the covariance is initialized to a value of $\boldsymbol{\Gamma}(0) < \rho_M I_p$.

Theorem 3. *Assume that the actual output can be represented as*

$$\mathbf{y}(t) = \boldsymbol{\phi}(t)^T\boldsymbol{\theta} \tag{83}$$

where $\boldsymbol{\phi}(t)$ is a bounded or unbounded vector and the transfer function (82) is strictly positive real. Then, the PAA (80) guarantees that

P1 . The parameter estimation error $\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}$ is uniformly bounded, i.e., $\tilde{\boldsymbol{\theta}}(t) \in L_\infty$. In addition, when $\mathbf{H}(p) = I_m$ in (81),

$$V_\theta(t) = \tilde{\boldsymbol{\theta}}(t)^T\boldsymbol{\Gamma}(t)^{-1}\tilde{\boldsymbol{\theta}}(t) \leq V_\theta(0) = \tilde{\boldsymbol{\theta}}(0)^T\boldsymbol{\Gamma}(0)^{-1}\tilde{\boldsymbol{\theta}}(0) \tag{84}$$

P2 . $\boldsymbol{\epsilon}, \boldsymbol{\nu} \in \mathcal{L}_2^m$, *i.e.,* $\lim_{t\to\infty} \int_0^t \|\boldsymbol{\epsilon}(\tau)\|^2 d\tau$ *and* $\lim_{t\to\infty} \int_0^t \|\boldsymbol{\nu}(\tau)\|^2 d\tau$ *exist. Thus, when the adaptation and prediction error vectors,* $\boldsymbol{\epsilon}$ *and* $\boldsymbol{\nu}$*, are uniformly continuous functions of time, they go to zero asymptotically as well:*

$$\lim_{t\to\infty} \boldsymbol{\nu}(t) = \lim_{t\to\infty} \boldsymbol{\epsilon}(t) = 0 \tag{85}$$

P3 . When $\boldsymbol{\phi}(t)$ *is uniformly bounded, i.e.,* $\boldsymbol{\phi} \in \mathcal{L}_\infty^{p\times m}$*, then,* $\dot{\hat{\boldsymbol{\theta}}} \in \mathcal{L}_2^p$*, i.e.,* $\lim_{t\to\infty} \int_0^t \|\dot{\hat{\boldsymbol{\theta}}}(\tau)\|^2 d\tau < \infty$. △

3 Parameter Identification in Dynamical Systems

In this section, we will focus on the parameter identification of several dynamic systems using the parameter estimation algorithms introduced in the previous section. From a practical point of view, plant model identification is not only a key starting point for any traditional control design techniques but also a major step toward the design of an appropriate adaptive controller and serves for the initialization of adaptive control systems.

3.1 Finite Impulse Response (FIR) Models

The forced response of an LTI system without disturbances is completely described by its unit impulse response $\{g(t), t \geq 0\}$ as

$$y(t) = \sum_{i=-\infty}^{t} g(t-i)u(i) = g(0)u(t) + g(1)u(t-1) + \cdots \tag{86}$$

For a stable system, $\{g(t), t \geq 0\}$ converges to zero exponentially fast and (86) may then be approximated by using only the first n terms in a practical application, which leads to the following FIR model (assuming $g(0) = 0$, i.e., at least one sampling delay in the system dynamics)

$$\hat{y}(t) = g(1)u(t-1) + g(2)u(t-2) + \cdots + g(n)u(t-n) \tag{87}$$

The FIR model (87) can be put into the linear regression form of (23) with

$$\begin{aligned} \boldsymbol{\varphi}(t)^T &= [u(t-1),\ u(t-2),\ \cdots,\ u(t-n)]^T \\ \boldsymbol{\theta} &= [g(1),\ g(2),\ \cdots,\ g(n)]^T \end{aligned} \tag{88}$$

and various parameter estimation algorithms introduced in section 2 can be used to obtain the estimates of the FIR model parameters, $\hat{\boldsymbol{\theta}}(t)$. For example, with the generalized RLSE algorithm in (57) and $\boldsymbol{\phi}(t) = \boldsymbol{\varphi}(t)$ and $\nu^0(t) = \epsilon^0(t) = y(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1)$, the estimation algorithm is given by

$$\begin{aligned}
\hat{\boldsymbol{\theta}}(t) &= \hat{\boldsymbol{\theta}}(t-1) + \tfrac{1}{1+\boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)}\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\epsilon^0(t) \\
\bar{\boldsymbol{\Gamma}}(t) &= \tfrac{1}{\lambda_1(t)}\left\{\boldsymbol{\Gamma}(t-1) - \tfrac{1}{\frac{\lambda_1(t)}{\lambda_2(t)}+\boldsymbol{\varphi}(t)^T\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)}\boldsymbol{\Gamma}(t-1)\boldsymbol{\varphi}(t)\boldsymbol{\varphi}^T(t)\boldsymbol{\Gamma}(t-1)\right\} \\
\boldsymbol{\Gamma}(t) &= \begin{cases} \boldsymbol{\Gamma}(t-1), & \text{if } \lambda_M\left(\bar{\boldsymbol{\Gamma}}(t)\right) > \rho_M \\ \bar{\boldsymbol{\Gamma}}(t) & \text{otherwise} \end{cases}
\end{aligned} \tag{89}$$

Furthermore, based on the analysis in subsections 2.3 and 2.4, we know that such an estimation will work well in implementation if the model truncation error from (86) to (87) is rather small or can be well approximated by a sequence of zero mean white noise sequence.

3.2 ARX Models

Consider the systems which can be described by the ARX model (10). With the linear regression form of predict output model (23) in which the regressor vector and the parameter vector are defined as in (8), the actual output of the system is in the form of (31) with the model error $e(t)$ being the same zero mean white noise sequence $d_e(t)$ as assumed in the ARX model (10). Thus the generalized RLSE algorithm (89) can be used to obtain an unbiased estimate of the ARX model parameters vector $\boldsymbol{\theta}$. This method for the identification of an LTI system is normally referred to as the ***equation error method*** as well.

3.3 Output Error Models

Consider the systems which can be described by the output error (OE) model (12) and characterized by the model parameter vector $\boldsymbol{\theta} = [l_1, \ldots, l_{n_l}, b_0, \ldots, b_{n_b}]^T \in R^{n_l+n_b+1}$. If the same way as in the parameter identification of ARX models is used to form the linear regression type predict output model, then the regressor would be

$$\boldsymbol{\varphi}(t)^T = [-y(t-1), \ldots, -y(t-n_l),\ u(t-d), \ldots, u(t-d-n_b)] \tag{90}$$

With this prediction model, the resulting model error $e(t)$ in (31) is related to $d_e(t)$ in the OE model (12) by

$$e(t) = L(q^{-1})d_e(t) \tag{91}$$

Thus though $d_e(t)$ is assumed to be a zero mean white noise sequence, $e(t)$ will not be white any more and an unbiased estimate of the OE model parameters vector $\boldsymbol{\theta}$ cannot be assured with the least square estimation algorithm in section 2. To bypass this problem, consider the fictitious situation where the following filtered output based on the input can be obtained first:

$$y_f(t) = \frac{q^{-d}B(q^{-1})}{L(q^{-1})}u(t) \tag{92}$$

i.e.,

$$y_f(t)+l_1y_f(t-1)+\cdots+l_{n_l}y_f(t-n_l) = b_0u(t-d)+\cdots+b_{n_b}u(t-d-n_b) \quad (93)$$

Then if the following regressor is used

$$\boldsymbol{\varphi}(t) = [-y_f(t-1),\ \cdots,\ -y_f(t-n_l),\ u(t-d),\ \cdots, u(t-d-n_b)]^T \quad (94)$$

the actual output of the system (12) will be in the form of (31) with the model error $e(t)$ being the same zero mean white noise sequence $d_e(t)$. The generalized RLSE algorithm (89) can then be used to obtain an unbiased estimate of the OE model parameters vector $\boldsymbol{\theta}$.

In reality, since the actual plant parameters are unknown, the filtered output in (93) cannot be obtained. As an alternative, the following filtered output $\hat{y}_f(t)$ based on the estimates of plant parameters at t, $\hat{\boldsymbol{\theta}}(t) = [\hat{l}_1(t),\ \ldots,\ \hat{l}_{n_l}(t),\ \hat{b}_0(t),\ \ldots, \hat{b}_{n_b}(t)]^T$, and the previous filtered outputs will be used instead of $y_f(t)$

$$\begin{aligned}\hat{y}_f(t) = -\hat{l}_1(t)\hat{y}_f(t-1) - \cdots - \hat{l}_{n_l}(t)\hat{y}_f(t-n_l)\\ +\hat{b}_0(t)u(t-d) + \cdots + \hat{b}_{n_b}(t)u(t-d-n_b)\end{aligned} \quad (95)$$

to form the regressor as

$$\boldsymbol{\varphi}(t) = [-\hat{y}_f(t-1),\ \cdots,\ -\hat{y}_f(t-n_l),\ u(t-d),\ \cdots, u(t-d-n_b)]^T \quad (96)$$

This method for the identification of an LTI system is normally referred to as the ***output error method***. However it should be noted that with this modification, even when $d_e(t) = 0, \forall t$, the actual output of the system (12) will not be in the form of (31) with $e(t) = 0$. Instead, it is straightforward to verify that it can be expressed as

$$y(t) = \boldsymbol{\varphi}(t)^T\boldsymbol{\theta} - q^{-1}L^*(q^{-1})\epsilon(t) + d_e(t) \quad (97)$$

where $L^*(q^{-1}) := l_1 + l_2q^{-1} + \cdots + l_{n_l}q^{-n_l+1} = q\left(L(q^{-1}) - 1\right)$ (i.e., $L(q^{-1}) = 1 + q^{-1}L^*(q^{-1})$) and $\epsilon(t)$ is the *a posteriori* prediction error defined in (48) as

$$\epsilon(t) = y(t) - \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t) \quad (98)$$

It is thus clear that the parameter estimates by the RLSE algorithm (89) is not guaranteed to converge even in the absence of plant disturbances (i.e., $d_e(t) = 0, \forall t$) unless certain conditions are met. See [5] for the stability conditions derived from the passivity theorems and further modifications in relaxing the restrictiveness of stability conditions.

3.4 ARMAX Models

Consider the systems which can be described by the ARMAX model (11) and characterized by the model parameter vector $\boldsymbol{\theta} = [a_1,\ \ldots,\ a_{n_a},\ b_0,\ \ldots, b_{n_b},$

$c_1, \ldots, c_{n_c}]^T \in R^{n_a+n_b+1+n_c}$. If the past disturbances, $d_e(j), j < t$, are known, then the ARMAX model (11) can be linearly parametrized by $\boldsymbol{\theta}$ by

$$\begin{aligned} y(t) &= -a_1 y(t-1) + \cdots - a_{n_a} y(t-n_a) + b_0 u(t-d) + \cdots \\ &\quad + b_{n_b} u(t-d-n_b) + d_e(t) + c_1 d_e(t-1) + \cdots + c_{n_c} d_e(t-n_c) \\ &= \boldsymbol{\varphi}(t)^T \boldsymbol{\theta} + d_e(t) \end{aligned} \tag{99}$$

where

$$\begin{aligned} \boldsymbol{\varphi}(t) = [&-y(t-1), \ldots, -y(t-n_a),\ u(t-d), \ldots, u(t-d-n_b), \\ &d_e(t-1), \ldots, d_e(t-n_c)]^T \end{aligned} \tag{100}$$

and the model error $e(t) = y(t) - \boldsymbol{\varphi}(t)^T \boldsymbol{\theta}$ would be the zero-mean white noise $d_e(t)$. In reality, since $d_e(j), j < t$ are unknown, the past prediction error $\epsilon(j) = y(j) - \boldsymbol{\varphi}(j)^T \hat{\boldsymbol{\theta}}(j)$ will be used as the estimate of $d_e(j)$ since $\epsilon(j) = d_e(j)$ when $\hat{\boldsymbol{\theta}}(j) = \boldsymbol{\theta}, \forall j$ (i.e., the ideal scenario when the parameter estimates are accurate). This leads to the following definition of regressor in practice

$$\begin{aligned} \boldsymbol{\varphi}(t) = [&-y(t-1), \ldots, -y(t-n_a),\ u(t-d), \ldots, u(t-d-n_b), \\ &\epsilon(t-1), \ldots, \epsilon(t-n_c)]^T \end{aligned} \tag{101}$$

This method for the identification of an LTI system is normally referred to as the ***extended least squares (ELS)*** algorithm.

With the regressor (101), the actual output of the system (11) can be expressed as

$$y(t) = \boldsymbol{\varphi}(t)^T \boldsymbol{\theta} - q^{-1} C^*(q^{-1}) \epsilon(t) + C(q^{-1}) d_e(t) \tag{102}$$

where $C^*(q^{-1}) := c_1 + c_2 q^{-1} + \cdots + c_{n_c} q^{-n_c+1} = q\left(C(q^{-1}) - 1\right)$ (i.e., $C(q^{-1}) = 1 + q^{-1} C^*(q^{-1})$). It is thus clear that the parameter estimates by the RLSE algorithm (89) is not guaranteed to converge even in the absence of plant disturbance of $d_e(t) = 0$ unless certain conditions are met as in the parameter identification of OE models.

3.5 Generalized Least Squares (GLS)

Consider the systems which can be described by the general LTI model (9) with $L = 1$ and characterized by the model parameter vector $\boldsymbol{\theta} = [a_1, \ldots, a_{n_a}, b_0, \ldots, b_{n_b}, c_1, \ldots, c_{n_c}, d_1, \ldots, d_{n_d}]^T \in R^{n_a+n_b+1+n_c+n_d}$. As in the ELS algorithm, if the past disturbances, $d_e(j), j < t$, are known, then such a model can be linearly parametrized by $\boldsymbol{\theta}$ and a zero mean white noise $d_e(t)$ as

$$\begin{aligned} y(t) = &-A^*(q^{-1}) y(t-1) + B(q^{-1}) u(t-d) + d_e(t) + C^*(q^{-1}) d_e(t-1) \\ &- D^*(q^{-1}) \left[A(q^{-1}) y(t-1) - B(q^{-1}) u(t-d-1)\right] \end{aligned} \tag{103}$$

where $A(q^{-1}) = 1+q^{-1}A^*(q^{-1})$ and $D(q^{-1}) = 1+q^{-1}D^*(q^{-1})$. This suggests the following the following adjustable predictor at t in practice

$$\hat{y}^0(t) = \boldsymbol{\varphi}(t)^T\hat{\boldsymbol{\theta}}(t-1) \tag{104}$$

where

$$\begin{aligned}\boldsymbol{\varphi}(t) = [&-y(t-1),\ldots,-y(t-n_a),\ u(t-d),\ldots,u(t-d-n_b),\\ &\epsilon(t-1),\ldots,\epsilon(t-n_c),\ \alpha(t-1),\ldots,\alpha(t-n_d)]^T\end{aligned} \tag{105}$$

in which $\epsilon(j) = y(j) - \boldsymbol{\varphi}(j)^T\hat{\boldsymbol{\theta}}(j)$ are the past *a posteriori* prediction errors and $\alpha(t)$ represents the estimate of $A(q^{-1})y(t) - B(q^{-1})u(t-d)$ with $\hat{\boldsymbol{\theta}}(t)$ (i.e., $\alpha(t-1) = \hat{A}(q^{-1},\hat{\boldsymbol{\theta}}(t-1))y(t-1) - \hat{B}(q^{-1},\hat{\boldsymbol{\theta}}(t-1))u(t-d-1)= y(t-1) + \hat{a}_1(t-1)y(t-2) + \cdots + \hat{a}_{n_a}(t-1)y(t-n_a-1) - \hat{b}_0(t-1)u(t-d-1) - \cdots - \hat{b}_{n_b}(t-1)u(t-d-n_b-1)$). This method for the identification of an LTI system is normally referred to as the ***generalized least squares (ELS)*** algorithm.

With the regressor (105), the actual output of the system (103) can be expressed as

$$\begin{aligned}y(t) = \boldsymbol{\varphi}(t)^T\boldsymbol{\theta} &- C^*(q^{-1})\epsilon(t-1) + C(q^{-1})d_e(t)\\ &-D^*(q^{-1})\left[\left(A^*(q^{-1} - \hat{A}^*(q^{-1},\hat{\boldsymbol{\theta}}(t-1))\right)y(t-1)\right.\\ &\left.-\left(B(q^{-1} - \hat{B}(q^{-1},\hat{\boldsymbol{\theta}}(t-1))\right)u(t-d-1)\right]\end{aligned} \tag{106}$$

It is thus clear that the parameter estimates by the RLSE algorithm (89) is not guaranteed to converge even in the absence of plant disturbance of $d_e(t) = 0$ unless certain conditions are met.

3.6 Nonlinear Models

The parameter estimation algorithms for linear regression type prediction model in previous section can also be used to identify parameters of certain nonlinear systems through the re-parameterization technique as well. The following example is used to illustrate this point.

Consider the nonlinear system described by

$$y(t) = c_1\left[y(t-1) + c_2y(t-1)u(t-1) + e^{c_3}u(t-1)\right] \tag{107}$$

where c_1, c_2, and c_3 are the three model parameters we would like to identify. As c_i do not appear linearly in the system dynamics, the previous estimation techniques cannot be direct applied. Instead, define the parameter vector as

$$\boldsymbol{\theta} = \begin{bmatrix} c_1 \\ c_1c_2 \\ c_1e^{c_3}\end{bmatrix} \tag{108}$$

Then the system dynamics (107) can be linearly parametrized as

$$y(t) = \boldsymbol{\varphi}(t)^T \boldsymbol{\theta} \tag{109}$$

where

$$\boldsymbol{\varphi}(t) = [y(t-1),\ y(t-1)u(t-1),\ u(t-1)]^T \tag{110}$$

It is thus clear that the RLSE algorithm (89) can be used to estimate $\boldsymbol{\theta}$, from which c_i can be backed out.

4 Experimental Conditions

As observed from example 1, convergence of parameter estimates depends not only on the type of estimation algorithm used (LSE is used in the example), but also on the type of input signals that are used in obtaining the experimental data, i.e., experimental conditions. As such, it is very important for one to know what type of signals will result in parameter convergence in general, which is the focus of this subsection. To gain better understanding of the issues, let us carefully examine the following examples first.

Example 3. Consider the system described by the following finite impulse response (FIR) model:

$$y(t) = b_0 u(t) + b_1 u(t-1) + e(t),\ \ t = 0, 1, \ldots \tag{111}$$

in which $\{e(t)\}$ is a sequence of independent normal distributed random variables of $N(0, \sigma)$. The system can be put in the form of (31) with

$$\begin{aligned} \boldsymbol{\varphi}(t)^T &= \left[\, u(t),\ \ u(t-1) \,\right] \\ \boldsymbol{\theta}^0 &= \left[\, b_0,\, b_1 \,\right]^T \end{aligned} \tag{112}$$

Thus, when the LSE algorithm without weighting is used, the resulting parameter estimates are given by (32) in which

$$\begin{aligned} \mathbf{P}(t)^{-1} = \boldsymbol{\Phi}(t)^T \boldsymbol{\Phi}(t) &= \sum_{i=1}^{i=t} \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T = \sum_{i=1}^{i=t} \begin{bmatrix} u(i) \\ u(i-1) \end{bmatrix} \left[\, u(i)\ u(i-1) \,\right] \\ &= \sum_{i=1}^{i=t} \begin{bmatrix} u(i)^2 & u(i)u(i-1) \\ u(i)u(i-1) & u(i-1)^2 \end{bmatrix} = \begin{bmatrix} \sum\limits_{i=1}^{i=t} u(i)^2 & \sum\limits_{i=1}^{i=t} u(i)u(i-1) \\ \sum\limits_{i=1}^{i=t} u(i)u(i-1) & \sum\limits_{i=1}^{i=t} u(i-1)^2 \end{bmatrix} \end{aligned} \tag{113}$$

The resulting parameter estimation errors $\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta}^0$ are then given by

$$\tilde{\boldsymbol{\theta}}(t) = \begin{bmatrix} \frac{1}{t}\sum\limits_{i=1}^{i=t} u^2(i) & \frac{1}{t}\sum\limits_{i=1}^{i=t} u(i)u(i-1) \\ \frac{1}{t}\sum\limits_{i=1}^{i=t} u(i)u(i-1) & \frac{1}{t}\sum\limits_{i=1}^{i=t} u^2(i-1) \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{t}\sum\limits_{i=1}^{i=t} u(i)e(i) \\ \frac{1}{t}\sum\limits_{i=1}^{i=t} u(i-1)e(i) \end{bmatrix} \tag{114}$$

Case 1: ***Input is a unit step***

Let us first consider the case when the experimental data for the above parameter estimation are obtained with a unit step input given by

$$u(t) = \begin{cases} 1 & t > 0 \\ 0 & t \leq 0 \end{cases} \tag{115}$$

Then the covariance matrix in (113) and the parameter estimation error (114) become

$$\begin{aligned} \mathbf{P}(t) &= \begin{bmatrix} t & t-1 \\ t-1 & t-1 \end{bmatrix}^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & \frac{t}{t-1} \end{bmatrix} \\ \tilde{\boldsymbol{\theta}}(t) &= \begin{bmatrix} 1 & \frac{t-1}{t} \\ \frac{t-1}{t} & \frac{t-1}{t} \end{bmatrix}^{-1} \begin{bmatrix} \frac{1}{t}\sum\limits_{i=1}^{i=t} e(i) \\ \frac{1}{t}\sum\limits_{i=2}^{i=t} e(i) \end{bmatrix} = \begin{bmatrix} e(1) \\ -e(1) + \frac{1}{t-1}\sum\limits_{i=2}^{i=t} e(i) \end{bmatrix} \end{aligned} \tag{116}$$

The following observations can then be made:

(i): As long as $t \geq 2$, $\boldsymbol{\Phi}(t)^T\boldsymbol{\Phi}(t)$ is nonsingular and $\mathbf{P}(t)$ is well defined. Thus, in the absence of model structure error (i.e., $e(i) = 0, \forall i$), the above LSE is unique and should be the same as the true value, as confirmed by (116).

(ii): Even with large data set (i.e., when $t \to \infty$), $\mathbf{P}(t)$ does not converge to zero, indicating that the LSE estimates may not converge to their true values when model errors exist (i.e., $e(i) \neq 0$ for some i). This is confirmed by the parameter estimation errors in (116), which approach to the non-zero values of $\tilde{\boldsymbol{\theta}}(\infty) = [e(1),\ -e(1)]^T$ for white noise $\{e(t)\}$.

(iii): Though in general the two LSE parameter estimates may not converge to their true values when the model structure errors exist, one linear combination of the estimates, $\hat{\theta}_1(t) + \hat{\theta}_2(t)$, does converge to its true value of $b_0 + b_1$ even when white noise model errors may exist. This is seen from (116) where

$$\tilde{\theta}_1(t) + \tilde{\theta}_2(t) = \frac{1}{t-1}\sum_{i=2}^{i=t} e(i) \quad \to \quad 0 \text{ as } t \to \infty \tag{117}$$

for white noises $\{e(t)\}$. To explain this phenomenon, note that the two eigenvalues of the co-variance matrix are

$$\lambda\left[\mathbf{P}(t)\right] = 1 + \tfrac{1}{2(t-1)} \pm \sqrt{1 + \tfrac{1}{4(t-1)^2}} \tag{118}$$

As $t \to \infty$, the co-variance matrix converges to a constant matrix of $\mathbf{P}(\infty)$ with the following eigenvalues and the associated eigenvectors

$$\begin{aligned} \lambda_1\left[\mathbf{P}(\infty)\right] &= 2, \quad \mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \end{bmatrix} \\ \lambda_2\left[\mathbf{P}(\infty)\right] &= 0, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} \end{aligned} \tag{119}$$

Thus, the parameter estimates converge along the direction of the eigenvector associated with $\lambda_2\left[P(\infty)\right] = 0$, i.e., $\mathbf{v}_2^T\boldsymbol{\theta} = b_0 + b_1$. For all parameter estimates to converge, it is necessary that all eigenvalues of $P(t)$ converge to zero.

This case illustrates that, with a step input, we can only identify at most one parameter for FIR models.

Case 2: ***Input u is a white noise independent of e***

In this case, u is assumed to be a zero mean white noise with $\mathrm{E}\left[u(t)^2\right] = \sigma_u$ and is independent of the model structure error e. Then, $R_u(0) = \bar{E}[u(t)^2] = \sigma_u$, $R_u(k) = \bar{E}[u(t)u(t-k)] = 0, \forall k \neq 0$, and $R_{ue}(j) = \bar{E}[u(t)e(t-j)] = 0, \forall j$. Thus, for large data set, from (113) and (114),

$$\begin{aligned} \tfrac{1}{t}\mathbf{P}(t)^{-1} &\overset{t\to\infty}{\longrightarrow} \begin{bmatrix} R_u(0) & R_u(1) \\ R_u(1) & R_u(0) \end{bmatrix} = \begin{bmatrix} \sigma_u & 0 \\ 0 & \sigma_u \end{bmatrix} \\ \tilde{\boldsymbol{\theta}}(t) &\overset{t\to\infty}{\longrightarrow} \begin{bmatrix} R_u(0) & R_u(1) \\ R_u(1) & R_u(0) \end{bmatrix}^{-1} \begin{bmatrix} R_{ue}(0) \\ R_{ue}(1) \end{bmatrix} = 0 \end{aligned} \tag{120}$$

which shows that the LSE estimates will converge to their true values even when the system may have zero mean white noise model errors. ♢

The above example illustrates that, in the presence of zero mean white noise model structure errors, for the LSE estimates to converge to their true values with large data sets, certain experimental conditions have to be met when obtaining the data sets. Those conditions are commonly referred to as the persistent excitation (PE) conditions and should be well-understood before a specific estimation algorithm is to be used. In the following subsections, the parameter estimation for systems escribed by finite impulse response (FIR) models will be examined in details first to illustrate what those conditions are and how to meet them.

4.1 Persistent Excitation (PE) Signals

Note that a FIR model is linearly parametrized by the parameter vector $\boldsymbol{\theta}$ in (88) with the regressor vector $\boldsymbol{\varphi}(t)$ uniquely determined by the input signal only, which significantly simplifies the convergence analysis of parameter estimation. Specifically, when the LSE algorithm (29) is used to obtain the n FIR model parameters experimentally, assuming identical weighting of $\mathbf{W}(t,i) = I$, the resulting parameter estimation error will be given by (35), in which $\mathbf{R}_{\boldsymbol{\varphi}}(0)$ is

$$\begin{aligned}\mathbf{R}_{\boldsymbol{\varphi}}(0) &= \bar{E}[\boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T] = \lim_{N\to\infty} \tfrac{1}{N}\sum_{t=1}^{N} E[\boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T] \\ &= \lim_{N\to\infty} \tfrac{1}{N}\sum_{t=1}^{N} E\left(\begin{bmatrix} u(t-1)^2 & \cdots & u(t-1)u(t-n) \\ u(t-2)u(t-1) & \cdots & u(t-2)u(t-n) \\ \vdots & \ddots & \vdots \\ u(t-n)u(t-1) & \cdots & u(t-n)^2 \end{bmatrix}\right) \\ &= \mathbf{C}_n \end{aligned} \tag{121}$$

where

$$\mathbf{C}_n := \begin{bmatrix} R_u(0) & R_u(1) & \cdots & R_u(n-2) & R_u(n-1) \\ R_u(1) & R_u(0) & \cdots & R_u(n-3) & R_u(n-2) \\ \vdots & \vdots & \ddots & \vdots & \vdots \\ R_u(n-2) & R_u(n-3) & \cdots & R_u(0) & R_u(1) \\ R_u(n-1) & R_u(n-2) & \cdots & R_u(1) & R_u(0) \end{bmatrix} \tag{122}$$

and $R_u(k) = \lim_{N\to\infty} \frac{1}{N}\sum_{t=1}^{N} E[u(t)u(t-k)]$.

Definition 1. *A signal u is called* ***weakly persistently exciting*** ***(PE) of order n*** *if the matrix $\mathbf{C}_n$ in (122) is positive definite.*

Definition 2. *A signal u is called* ***strongly persistent exciting of order n*** *if there exist $\rho_1 \geq \rho_2 > 0$ and an integer l such that*

$$\rho_1 I_n \;\geq \sum_{i=t-l}^{t} \boldsymbol{\varphi}(i)\boldsymbol{\varphi}(i)^T \geq \rho_2 I_n, \quad \forall t \geq t_0 \tag{123}$$

for some t_0.

Note that (123) is the same as

$$\rho_1 I_n \geq \begin{bmatrix} \sum_{i=t-l}^{t} u(i-1)^2 & \cdots & \sum_{i=t-l}^{t} u(i-1)u(i-n) \\ \vdots & \ddots & \vdots \\ \sum_{i=t-l}^{t} u(i-n)u(i-1) & \cdots & \sum_{i=t-l}^{t} u(i-n)^2 \end{bmatrix} \geq \rho_2 I_n \tag{124}$$

It is noted that the strongly PE order of n of a signal implies the weakly PE order of n, and the PE order of n of a signal is also PE order of k for any $k \leq n$ as well. Furthermore, from the analysis on the convergence properties of LSE in subsections 2.3 and 2.4, it is clear that the following conclusions can be drawn:

i For the FIR model with an order of p, to be able to identify all p model parameters in the presence of zero mean white noise model errors, it is necessary to collect the data with an input of weakly PE order of at least p when the LSE algorithm with non-diminishing weightings (e.g., identical weightings) is used. With this experimental condition, the variance of LSE parameter estimation error converges to zero at a rate of $1/t$.

The strongly PE condition is needed for on-line parameter estimations when a forgetting factor (or diminishing weighting) has to be used to discard old data to be able to track slowly changing parameters or the sudden change of model parameters in the middle of operations. When a diminishing weighting or forgetting factor is used, the input needs to be strongly PE order of at least p to identify all p model parameters of a FIR model.

Theorem 4. *A signal u is weakly persistently exciting of order n if and only if (iff)*

$$U = \lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} \left(A(q)u(k)\right)^2 \ > 0 \tag{125}$$

for all nonzero polynomials A of degree $n-1$ or less. △

Example 4. ***Pulse inputs are not PE order of any.*** ♢

Example 5. ***Step inputs are PE order of 1 only***
Consider a unit step input of

$$u_s(t) = \begin{cases} 1 & t \geq 0 \\ 0 & t < 0 \end{cases} \tag{126}$$

For the polynomial $A(q) = q - 1$,

$$A(q)u_s(k) = (q-1)u_s(k) = u_s(k+1) - u_s(k) = 0, \quad \forall k \geq 0 \tag{127}$$

Thus, from Theorem 4, the unit step input u_s cannot be PE order of 2 or higher. On the other hand, since

$$C_1 = [R_u(0)] = \lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} u_s(k)^2 = 1 \tag{128}$$

u_s is PE order of 1. ♢

Example 6. ***Periodic inputs***
Consider a periodic input $u(t)$ with period N, i.e., $u(t+N) = u(t), \forall t$. Then,

$$\left(q^N - 1\right) u(t) = 0 \tag{129}$$

Thus the signal can at most be PE order of N. ♢

Theorem 5. *A stationary input $\{u(t)\}$ is weakly persistently exciting of order n if its two sided spectrum is nonzero at n points (or more).* △

Theorem 5 implies that

i. A sinusoidal input of a single frequency, $\{u(t) = A_u sin(\omega_1 t)\}$, is weakly persistently exciting of order 2.
ii. A linear combination k sinusoidal signals having different frequencies, $\{u(t) = \sum_{i=1}^{k} A_{ui} sin(\omega_i t),\ \omega_1 \neq \cdots \neq \omega_k\}$, is weakly persistently exciting of order $2k$.
iii. A square wave input of period N is weakly persistently exciting of order N.
iv. A gaussian white noise input is weakly persistently exciting of any order.

Theorem 6. *Assume that u is weakly persistently exciting of order n and $B(q)$ and $A(q)$ are some* non-zero *polynomials. Then,*

(a) The filtered signal v defined by

$$v(t) = B(q)u(t) \tag{130}$$

is weakly persistently exciting of order l for some $l \geq n - m$ where $m < n$ is the degree of $B(q)$.

(b) If $A(q)$ has all roots inside the unit circle (i.e., a Hurwitz polynomial), the signal w generated by

$$w(t) = \frac{1}{A(q)} u(t) \tag{131}$$

is persistently exciting of order n. △

4.2 Order of Transfer Function Models

The following example illustrates that, to identify parameters of a dynamics systems described by a transfer function, it is normally necessary to know the order of the transfer function. Otherwise, we may have the problem of losing identifiability, leading to over-fitting and erroneous parameter estimates.

Example 7. Consider a system described by

$$y(t) + a_1^0 y(t-1) = b_0^0 u(t-1) + e(t) \tag{132}$$

Let us assume that we do not know the order of the actual process (132) (i.e., $n_{A^0} = 1$, $d^0 = 1$, and $n_{B^0} = 0$, and try to use a design model given by

$$A(q^{-1})y(t) = q^{-d} B(q^{-1})u(t) \tag{133}$$

to fit the data. For simplicity, assume $n_a = 2$, $d = 1$, and $n_b = 1$ to be used. Then, the regression model resulting from the design model (133) is

$$\begin{aligned} &y(t) + a_1 y(t-1) + a_2 y(t-2) = b_0 u(t-1) + b_1 u(t-2) \\ &y(t) = \boldsymbol{\varphi}(t)^T \boldsymbol{\theta} \end{aligned} \tag{134}$$

where

$$\begin{aligned} &\boldsymbol{\theta} = [a_1,\ a_2,\ b_0,\ b_1]^T \\ &\boldsymbol{\varphi}(t) = [-y(t-1),\ -y(t-2),\ u(t-1),\ u(t-2)]^T \end{aligned} \tag{135}$$

For simplicity, assume that $e(t) = 0, \forall t$ in (132). Then, no matter what type of inputs we supplied to system to collect the experimental data for parameter estimation, we always have

$$y(t) = -a_1^0 y(t-1) + b_0^0 u(t-1) \tag{136}$$

Thus, $\forall t$, the first column of $\boldsymbol{\Phi}(t)$ will always be a linear combination of its second and the fourth column, and the resulting matrix $\boldsymbol{\Phi}(t)^T \boldsymbol{\Phi}(\mathbf{t})$ is singular. So the LSE estimates are not unique and RLSE estimates will never converge to their true values no matter what type of input signals we can choose. $\diamondsuit$

When the order of the actual process is known, the following theorem gives conditions for the convergence of LSE parameter estimates to their true values.

Theorem 7. *Consider data generated by an actual process described by*

$$\begin{aligned} A^0(q)y(t) = B^0(q)u(t) + e(t),\ \ A^0(q) &= q^{n_{A^0}} + a_1^0 q^{n_{A^0}-1} + \cdots + a_{n_{A^0}}^0 \\ B^0(q) &= b_0^0 q^{n_{B^0}} + b_1^0 q^{n_{B^0}-1} + \cdots + b_{n_{B^0}}^0 \end{aligned} \tag{137}$$

with $A^0(q)$ stable, $A^0(q)$ and $B^0(q)$ being co-prime, and $e(t) = 0$. Suppose that the orders of the system are known so that the following assumed model

$$\begin{aligned} A(q)y(t) = B(q)u(t),\ \ A(q) &= q^{n_a} + a_1 q^{n_a-1} + \cdots + a_{n_a} \\ B(q) &= b_0 q^{n_b} + b_1 q^{n_b-1} + \cdots + b_{n_b} \end{aligned} \tag{138}$$

with $n_a = n_{A^0}$ and $n_b \geq n_{B^0}$ is used to identify the system parameters, i.e., consider a linear regression model with the regressor and the parameter vector given by

$$\begin{aligned} &\boldsymbol{\theta} = [a_1, \cdots, a_{n_a},\ b_0, \cdots, b_{n_b}]^T \\ &\boldsymbol{\varphi}(t) = [-y(t-1), \cdots, -y(t-n_a),\ u(t-n_a+n_b), \cdots, u(t-n_a)]^T \end{aligned} \tag{139}$$

Then, if u is PE of order $p = n_a + n_b + 1$,

$$\lim_{t\to\infty} \tfrac{1}{t} \boldsymbol{\Phi}(t)^T \boldsymbol{\Phi}(t) > 0 \tag{140}$$

and the LSE $\hat{\boldsymbol{\theta}}(t)$ converge to their true values of $\boldsymbol{\theta} = [\boldsymbol{\theta}^{0^T}, 0, \cdots, 0]^T$ where $\boldsymbol{\theta}^0 = [a_1^0, \cdots a_{n_{A^0}}^0,\ b_0^0, \cdots b_{n_{B^0}}^0]^T$. $\triangle$

Remark 3. In the above theorem, when $\{e(t)\}$ in (137) is a white noise sequence, then, for models of all orders of $n_a = n_{A^0}$, (140) is true as long as the input u is PE order of $n_b + 1$ only. $\diamondsuit$

4.3 Persistent Excitation Conditions in Adaptive Control

In adaptive control, persistent excitation (PE) conditions for the convergence of on-line parameter estimates to their true values are normally defined in terms of the regressor vector instead of the input signal as the regressor contains other signals as well. Furthermore, as certain forgetting factors or some equivalent means (e.g., the use of constant non-zero adaptation rate all the time in the gradient type of adaptation law) have to be used to keep the parameter estimation alive in order to track slowly changing parameters, conditions similar to the strongly PE condition of an input signal in (137) have to be introduced, as detailed in the following theorems.

Theorem 8. *Consider the general PAA given by (57). When the following PE condition is satisfied,*

$$\begin{aligned} &\exists\ \rho_2 > \rho_1 > 0, \,\&\, l, t_0 \in \mathcal{Z} \quad s.t. \\ &\qquad \rho_1 I_p \leq \sum_{i=t-l}^{t} \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \leq \rho_2 I_p,\ \forall t \geq t_0 \end{aligned} \tag{141}$$

then, in addition to the results in Theorem 1,

P4 . $\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta} \in l_2$*, and the parameter estimates converge to their true values asymptotically, i.e.,* $\tilde{\boldsymbol{\theta}}(t) = \hat{\boldsymbol{\theta}}(t) - \boldsymbol{\theta} \to 0$ *as* $t \to \infty$*.* △

Remark 4. Noting that the second inequality of the PE condition (141) is that same as that the observation vector is uniformly bounded, i.e., $\boldsymbol{\phi} \in l_\infty^p$. ♢

The corresponding theorem in continuous time domain is as follows.

Theorem 9. *Consider the general PAA given by (80). When* $\boldsymbol{\phi} \in \mathcal{L}_\infty^p$*, i.e.,* $\|\boldsymbol{\phi}(t)\| \leq \rho_\phi$ *for some* $\rho_\phi > 0$*, and the following PE condition is satisfied,*

$$\begin{aligned} &\exists\ \rho > 0, T > 0, t_0 \in \mathcal{R} \quad s.t. \\ &\qquad \tfrac{1}{T}\int_{t-T}^{t} \boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T d\tau \geq \rho I_p,\ \forall t \geq t_0 \end{aligned} \tag{142}$$

then, in addition to the results in Theorem 3,

P4 . $\tilde{\boldsymbol{\theta}} \in \mathcal{L}_2^p$*, and the parameter estimates converge to their true values asymptotically, i.e.,* $\tilde{\boldsymbol{\theta}}(t) \to 0$ *as* $t \to \infty$*.* △

Remark 5. The PE condition (142) is equivalent to

$$\begin{aligned} &\exists\ \alpha > 0,\ T > 0, t_0 \in \mathcal{R} \ \text{ s.t. } \forall \mathbf{q} \in \mathcal{R}^n \text{ with } \|\mathbf{q}\| = 1 \\ &\tfrac{1}{T}\int_{t-T}^{t} \left(\mathbf{q}^T \boldsymbol{\phi}(\tau)\right)^2 d\tau \geq \alpha, \quad \forall t \geq t_0 \end{aligned} \tag{143}$$

Thus, the PE condition can be interpreted as a condition on the energy of $\boldsymbol{\phi}(t)$ in all directions. ♢

4.4 Input Design for Open-Loop Experiments

As revealed in previous analysis, the choice of input signals in obtaining the experimental data needed for system identification plays an important role in the quality of resulting parameter estimates. The selection of input signals in practice should at least consider the following factors:

a **Informative enough**. The input signal needs to be persistently exciting of order high enough to be able to identify all concerned model parameters in principle.

b **Maximizing signal-to-noise-ratio**. The input signals determine the operating point of the system and which parts and modes of the system are excited during the experiment, which has a very substantial influence on the signal-to-noise-ratio of the collected experimental data. The input signals need to be chosen to maximize the signal-to-noise-ratio to improve the estimation accuracy (e.g., reducing model errors by using input signals not violating the assumptions of the assumed model structure).

c **Minimizing variance**. In general, a small variance in the estimation of a certain component of the parameter vector is obtained when the predictor is sensitive to that component. Hence, when possible, one should choose the outputs $\mathbf{y}(t)$ to fit (need not be the same as the measured process outputs) and the inputs $u(t)$ so that the predicted output for parameter estimation becomes sensitive with respect to parameters which are important to the application in question.

With the above general guideline in mind, one may still have a substantial amount of freedom for the actual choice of input signals. For linear system identification, in practice, one may first decide the frequency band to identify the system in question, and then select a signal with a more or less flat spectrum over this band. Some typical choices of input signals are provided in the MATLAB System Identification Toolbox with command "**idinput**", where a comprehensive estimation and analysis GUI, "**ident**" is also provided.

References

1. K. J. Astrom. *Introduction to Stochastic Control Theory.* Academic Press, New York, 1970.
2. K. J. Astrom and B. Wittenmark. *Adaptive control.* Addison-wesley Publishing Company, (second edition), 1995.
3. G. C. Goodwin and Shin. *Adaptive Filtering Prediction and Control.* Englewood Cliffs, New Jersey, Prentice-Hall, 1984.
4. R. Isermann, K. H. Lachmann, and D. Matko. *Adaptive Control Systems.* Prentice Hall, New York, 1992.
5. I. D. Landau, R. Lozano, and M. M'Saad. *Adaptive control.* Springer, New York, 1998.

6. L. Ljung. *System Identification – Theory for the User*. Prentice-Hall, Inc., Upper Saddle River, NJ 07458, USA, 1999.
7. K. S. Narendra and A. M. Annaswamy. A new adaptive law for robust adaptive control without persistent excitation. *IEEE Trans. on Automatic Control*, 32:134–145, 1987.
8. M. Tomizuka. Parallel mras without compensation block. *IEEE Trans. on Automatic Control*, 27(2):505–506, 1982.
9. M. Tomizuka. On a relaxation of spr condition in parallel mras—continuous time case. *IEEE Trans. on Automatic Control*, 33(10):976–979, 1988.
10. M. Tomizuka and Y. Takahashi. Deterministic disturbance rejection in linear identification. *Trans. of ASME, Journal of of Dynamic Systems, Measurement and Control*, 99(4):307–310, 1977.

Appendix: Proofs of All Lemmas and Theorems

Proof. **Proof of Lemma 1**
First note that with $n_c \leq n_a + j - 1$, there always exists a unique $J(q^{-1})$ and $F_j(q^{-1})$ of the given form satisfying (19). Multiplying the ARMAX model (11) by $J(q^{-1})q^j$ and applying (19):

$$\begin{aligned}\left[C(q^{-1}) - q^{-j}F_j(q^{-1})\right] y(t+j) = B(q^{-1})J(q^{-1})u(t-d+j) \\ +C(q^{-1})J(q^{-1})d_e(t+j)\end{aligned} \tag{144}$$

which leads to

$$y(t+j) = \tfrac{F_j(q^{-1})}{C(q^{-1})}y(t) + \tfrac{B(q^{-1})J(q^{-1})}{C(q^{-1})}u(t-d+j) + J(q^{-1})d_e(t+j) \tag{145}$$

Thus,

$$\begin{aligned}&E\left\{[y(t+j) - \hat{y}(t+j \mid t)]^2\right\} \\ &\quad = E\left\{[\tfrac{F_j(q^{-1})}{C(q^{-1})}y(t) + \tfrac{B(q^{-1})J(q^{-1})}{C(q^{-1})}u(t-d+j) - \hat{y}(t+j \mid t)]^2\right\} \\ &\quad +E\left\{[\tfrac{F_j(q^{-1})}{C(q^{-1})}y(t) + \tfrac{B(q^{-1})J(q^{-1})}{C(q^{-1})}u(t-d+j) - \hat{y}(t+j \mid t)][J(q^{-1})d_e(t+j)]\right\} \\ &\quad +E\left\{[J(q^{-1})d_e(t+j)]^2\right\}\end{aligned} \tag{146}$$

Noting the predictor structure (14) and the fact that $J(q^{-1})d_e(t+j)$ consists of future noises $d_e(t+1), \ldots, d_e(t+j)$ only, the second term in the right hand side of (146) is zero. Therefore, the optimal predictor should be the one which makes the first term zero, which is given by (18). This proves the lemma. ♢

Proof. **Proof of Lemma 2**
By direct substitution, it is easy to verify that

$$\begin{aligned}&(\mathbf{A} + \mathbf{BCD})\left[\mathbf{A}^{-1} - \mathbf{A}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1}\right] \\ &\quad = \mathbf{I} - \mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1} + \mathbf{BCDA}^{-1} \\ &\qquad - \mathbf{BCDA}^{-1}\mathbf{B}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1} \\ &\quad = \mathbf{I} + \mathbf{BC}\left\{-\mathbf{C}^{-1} + \left(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B}\right) - \mathbf{DA}^{-1}\mathbf{B}\right\}(\mathbf{C}^{-1} + \mathbf{DA}^{-1}\mathbf{B})^{-1}\mathbf{DA}^{-1} \\ &\quad = \mathbf{I}\end{aligned} \tag{147}$$

Therefore, $\mathbf{A} + \mathbf{BCD}$ is invertible and (40) is true. ♢

Proof. **Proof of Theorem 1**
The proof of the theorem for SISO case without covariance limiting is given in [5] using the passivity (or hyperstability) approach. The proof of the theorem with the covariance limiting as stated in (57) is detailed below.

Let $(\mathbf{A}_H,\ \mathbf{B}_H,\ \mathbf{C}_H,\ \mathbf{D}_H)$ be the state-space matrices of a minimal realization for the transfer function matrix $\mathbf{H}(z^{-1})$. Then the relationship (58) can be represented in state-space by

$$\begin{aligned}\mathbf{x}_H(t+1) &= \mathbf{A}_H\mathbf{x}_H(t) + \mathbf{B}_H\boldsymbol{\epsilon}(t) \\ \boldsymbol{\nu}(t) &= \mathbf{C}_H\mathbf{x}_H(t) + \mathbf{D}_H\boldsymbol{\epsilon}(t)\end{aligned} \tag{148}$$

where $\mathbf{x}_H(t) \in \mathcal{R}^{n_H}$ is the state vector of the minimal realization. Viewing (59), $(\mathbf{A}_H,\ \mathbf{B}_H,\ \mathbf{C}_H,\ \mathbf{D}_H - \frac{\lambda_H}{2} I_m)$ are the state-space matrices of a minimal realization for the transfer function matrix $\mathbf{H}'(z^{-1})$. As $\mathbf{H}'(z^{-1})$ is SPR by assumption, from KYP lemma, there exist $\mathbf{P}_H > 0$, $\mathbf{Q}_H > 0$, and $\mathbf{L}_H$ and $\mathbf{K}_H$ such that

$$\begin{aligned}
&\mathbf{A}_H^T \mathbf{P}_H \mathbf{A}_H - \mathbf{P}_H = -\mathbf{L}_H \mathbf{L}_H^T - \mathbf{Q}_H \\
&\mathbf{B}_H^T \mathbf{P}_H \mathbf{A}_H + \mathbf{K}_H^T \mathbf{L}_H^T = \mathbf{C}_H \\
&\mathbf{K}_H^T \mathbf{K}_H = \mathbf{D}_H - \tfrac{\lambda_H}{2} I_m + \left(\mathbf{D}_H - \tfrac{\lambda_H}{2} I_m\right)^T - \mathbf{B}_H^T \mathbf{P}_H \mathbf{B}_H
\end{aligned} \tag{149}$$

Define

$$V_H(t) = \mathbf{x}_H(t)^T \mathbf{P}_H \mathbf{x}_H(t) \tag{150}$$

Then, from (148) and (149),

$$\begin{aligned}
\Delta V_H\ (t+1) &= \mathbf{x}_H(t+1)^T \mathbf{P}_H \mathbf{x}_H(t+1) - \mathbf{x}_H(t)^T \mathbf{P}_H \mathbf{x}_H(t) \\
&= [\mathbf{A}_H \mathbf{x}_H(t) + \mathbf{B}_H \epsilon(t)]^T \mathbf{P}_H [\mathbf{A}_H \mathbf{x}_H(t) + \mathbf{B}_H \epsilon(t)] - \mathbf{x}_H(t)^T \mathbf{P}_H \mathbf{x}_H(t) \\
&= \mathbf{x}_H(t)^T \left[\mathbf{A}_H^T \mathbf{P}_H \mathbf{A}_H - \mathbf{P}_H\right] \mathbf{x}_H(t) + 2\boldsymbol{\epsilon}(t)^T \mathbf{B}_H^T \mathbf{P}_H \mathbf{A}_H \mathbf{x}_H(t) \\
&\quad + \boldsymbol{\epsilon}(t)^T \mathbf{B}_H^T \mathbf{P}_H \mathbf{B}_H \boldsymbol{\epsilon}(t) \\
&= \mathbf{x}_H(t)^T \left[-\mathbf{L}_H \mathbf{L}_H^T - \mathbf{Q}_H\right] \mathbf{x}_H(t) + 2\boldsymbol{\epsilon}(t)^T [\mathbf{C}_H - \mathbf{K}_H^T \mathbf{L}_H^T] \mathbf{x}_H(t) \\
&\quad + \boldsymbol{\epsilon}(t)^T [\mathbf{D}_H + \mathbf{D}_H^T - \lambda_H I_m - \mathbf{K}_H^T \mathbf{K}_H] \boldsymbol{\epsilon}(t) \\
&= -\mathbf{x}_H(t)^T \mathbf{Q}_H \mathbf{x}_H(t) - \|\mathbf{L}_H^T \mathbf{x}_H(t) + \mathbf{K}_H \epsilon(t)\|^2 + 2\boldsymbol{\epsilon}(t)^T \boldsymbol{\nu}(t) - \lambda_H \|\boldsymbol{\epsilon}(t)\|^2
\end{aligned} \tag{151}$$

Thus, $\forall t$

$$2\boldsymbol{\epsilon}(t)^T \boldsymbol{\nu}(t) \geq \Delta V_H(t+1) + \lambda_H \|\boldsymbol{\epsilon}(t)\|^2 + \mathbf{x}_H(t)^T \mathbf{Q}_H \mathbf{x}_H(t) \tag{152}$$

On the other hand, $V_\theta(t)$ in (61) satisfies

$$\begin{aligned}
\Delta V_\theta(t) &= \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t)^{-1} \tilde{\boldsymbol{\theta}}(t) - \tilde{\boldsymbol{\theta}}(t-1)^T \boldsymbol{\Gamma}(t-1)^{-1} \tilde{\boldsymbol{\theta}}(t-1) \\
&= \underbrace{\tilde{\boldsymbol{\theta}}(t)^T \left[\boldsymbol{\Gamma}(t)^{-1} - \boldsymbol{\Gamma}(t-1)^{-1}\right] \tilde{\boldsymbol{\theta}}(t)}_{I_V} \\
&\quad + \underbrace{\tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t-1)^{-1} \tilde{\boldsymbol{\theta}}(t) - \tilde{\boldsymbol{\theta}}(t-1)^T \boldsymbol{\Gamma}(t-1)^{-1} \tilde{\boldsymbol{\theta}}(t-1)}_{II_V}
\end{aligned} \tag{153}$$

From (57), when $\lambda_M\left(\bar{\boldsymbol{\Gamma}}(t)\right) > \rho_M$, $\boldsymbol{\Gamma}(t)^{-1} = \boldsymbol{\Gamma}(t-1)^{-1}$ and I_V in (153) becomes zero. When $\lambda_M\left(\bar{\boldsymbol{\Gamma}}(t)\right) \leq \rho_M$, $\boldsymbol{\Gamma}(t)^{-1} = \bar{\boldsymbol{\Gamma}}(t)^{-1}$ and I_V in (153) becomes

$$\begin{aligned}
I_V &= \tilde{\boldsymbol{\theta}}(t)^T \left[(\lambda_1(t) - 1)\, \boldsymbol{\Gamma}(t-1)^{-1} + \lambda_2(t) \boldsymbol{\phi}(t) \boldsymbol{\phi}(t)^T\right] \tilde{\boldsymbol{\theta}}(t) \\
&= -\left[1 - \lambda_1(t)\right] \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t-1)^{-1} \tilde{\boldsymbol{\theta}}(t) + \lambda_2(t) \|\boldsymbol{\epsilon}(t)\|^2 \\
&\leq \lambda_{2M} \|\boldsymbol{\epsilon}(t)\|^2
\end{aligned} \tag{154}$$

From the first equation of (57),

$$\tilde{\boldsymbol{\theta}}(t) = \tilde{\boldsymbol{\theta}}(t-1) + \boldsymbol{\Gamma}(t-1) \boldsymbol{\phi}(t) \boldsymbol{\nu}(t) \tag{155}$$

Thus II_V in (153) becomes

$$\begin{aligned} II_V &= \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t-1)^{-1}\tilde{\boldsymbol{\theta}}(t) - \tilde{\boldsymbol{\theta}}(t-1)^T \left[\boldsymbol{\Gamma}(t-1)^{-1}\tilde{\boldsymbol{\theta}}(t) - \boldsymbol{\phi}(t)\boldsymbol{\nu}(t)\right] \\ &= \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\Gamma}(t-1)^{-1}\left[\tilde{\boldsymbol{\theta}}(t) - \tilde{\boldsymbol{\theta}}(t-1)\right] + \tilde{\boldsymbol{\theta}}(t-1)^T \boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &= \tilde{\boldsymbol{\theta}}(t)^T \boldsymbol{\phi}(t)\nu(t) + \tilde{\boldsymbol{\theta}}(t-1)^T \boldsymbol{\phi}(t)\nu(t) \\ &= -\boldsymbol{\epsilon}(t)^T \boldsymbol{\nu}(t) + [\tilde{\boldsymbol{\theta}}(t) - \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t)]^T \boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &= -2\boldsymbol{\epsilon}(t)\boldsymbol{\nu}(t) - \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \end{aligned} \tag{156}$$

Substituting (155) and (156) into (153) and noting (152),

$$\begin{aligned} \Delta V_\theta(t) &\le \lambda_{2M}\|\boldsymbol{\epsilon}(t)\|^2 - 2\boldsymbol{\epsilon}(t)\boldsymbol{\nu}(t) - \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &\le -\lambda_\epsilon\|\boldsymbol{\epsilon}(t)\|^2 - \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) - \Delta V_H(t+1) - \lambda_{min}\left(\mathbf{Q}_H\right)\|\mathbf{x}_H(t)\|^2 \end{aligned} \tag{157}$$

where $\lambda_\epsilon = \lambda_{2H} - \lambda_{2M} > 0$. As (157) is valid for any t, it is clear that

$$\begin{aligned} \lambda_\epsilon \textstyle\sum_{i=1}^t \|\boldsymbol{\epsilon}(i)\|^2 &+ \lambda_{min}\left(\mathbf{Q}_H\right)\textstyle\sum_{i=1}^t \|\mathbf{x}_H(i)\|^2 + \sum_{i=1}^t \boldsymbol{\nu}(i)^T \boldsymbol{\phi}(i)^T \boldsymbol{\Gamma}(i-1)\boldsymbol{\phi}(i)\nu(i) \\ &\le -\textstyle\sum_{i=1}^t \left[\Delta V_\theta(i) + \Delta V_H(i+1)\right] \\ &= V_\theta(0) - V_\theta(t) + V_H(1) - V_H(t+1) \\ &= V_\theta(0) + V_H(1) \end{aligned} \tag{158}$$

Thus $\boldsymbol{\epsilon} \in l_2^m, \mathbf{x}_H \in l_2^{n_H}$, and $\sqrt{\boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)}\boldsymbol{\nu}(t) \in l_2^m$. Noting (148), $\boldsymbol{\nu} \in l_2^m$ as well, which proves P2 of the theorem.

Noting $\Gamma(t) \le \rho_M I_p$ by the co-variance limiting, from (158),

$$\tfrac{1}{\rho_M}\|\tilde{\boldsymbol{\theta}}(t)\|^2 \le V_\theta(t) \le V_\theta(0) + V_H(1) \tag{159}$$

which shows that $\tilde{\boldsymbol{\theta}}(t)$ is uniformly bounded. Furthermore, when $H = 1$, there is no state in (148) (i.e., $n_H = 0$) and $V_H(t) = 0$. Thus (61) is true.

To prove P3, noting (57),

$$\begin{aligned} &\hat{\boldsymbol{\theta}}(t) - \hat{\boldsymbol{\theta}}(t-1) = \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &\|\hat{\boldsymbol{\theta}}(t) - \hat{\boldsymbol{\theta}}(t-1)\|^2 = \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)^2 \boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &\qquad = \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)^{1/2}\boldsymbol{\Gamma}(t-1)\boldsymbol{\Gamma}(t-1)^{1/2}\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &\qquad \le \lambda_M\left(\boldsymbol{\Gamma}(t-1)\right)\boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \\ &\qquad \le \rho_M \boldsymbol{\nu}(t)^T \boldsymbol{\phi}(t)^T \boldsymbol{\Gamma}(t-1)\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) \end{aligned} \tag{160}$$

Thus, noting (63), P3 of the theorem (64) is true, which completes the proof. ♢

Proof. **Proof of Theorem 2**
By the definition of the weighting in (77), it is easy to verify that

$$\tfrac{\partial \mathbf{W}(t,\tau)}{\partial t} = e^{-\int_\tau^t \mu(s)ds}\left[-\mu(t)\right] = -\mu(t)\mathbf{W}(t,\tau) \tag{161}$$

Differentiating (75):

$$\begin{aligned}\dot{\mathbf{R}}(t)\hat{\boldsymbol{\theta}}(t)+\mathbf{R}(t)\dot{\hat{\boldsymbol{\theta}}}(t) &= \int_0^t \boldsymbol{\varphi}(\tau)\tfrac{\partial \mathbf{W}(t,\tau)}{\partial t}\mathbf{y}(\tau)d\tau+\boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\mathbf{y}(\tau)|_{\tau=t}\\ &= -\mu(t)\int_0^t \boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\mathbf{y}(\tau)d\tau+\boldsymbol{\varphi}(t)\mathbf{y}(t)\\ &= -\mu(t)\mathbf{R}(t)\hat{\boldsymbol{\theta}}(t)+\boldsymbol{\varphi}(t)\mathbf{y}(t)\end{aligned} \tag{162}$$

in which (161) has been used. On the other hand, by the definition of $\mathbf{R}(t)$,

$$\begin{aligned}\dot{\mathbf{R}}(t) &= \int_0^t \boldsymbol{\varphi}(\tau)\tfrac{\partial \mathbf{W}(t,\tau)}{\partial t}\boldsymbol{\varphi}(\tau)^T d\tau+\boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\boldsymbol{\varphi}(\tau)^T\big|_{\tau=t}\\ &= -\mu(t)\int_0^t \boldsymbol{\varphi}(\tau)\mathbf{W}(t,\tau)\boldsymbol{\varphi}(\tau)^T d\tau+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T\\ &= -\mu(t)\mathbf{R}(t)+\boldsymbol{\varphi}(t)\boldsymbol{\varphi}(t)^T\end{aligned} \tag{163}$$

Substituting (163) into (162) and noting $\mathbf{P}(t)=\mathbf{R}^{-1}(t)$, the first equation of (78) is proved.

Taking the derivative of $\mathbf{P}(t)\mathbf{R}(t)=I_p$

$$\dot{\mathbf{P}}(t)\mathbf{R}(t)+\mathbf{P}(t)\dot{\mathbf{R}}(t)=0 \tag{164}$$

which leads to

$$\dot{\mathbf{P}}(t)=-\mathbf{P}(t)\dot{\mathbf{R}}(t)\mathbf{R}^{-1}(t) \tag{165}$$

Substituting (163) into (165) and noting $\mathbf{P}(t)=\mathbf{R}^{-1}(t)$, the second equation of (78) is proved. ♢

Proof. **Proof of Theorem 3**
First note that when (83) is true, from (81),

$$\boldsymbol{\epsilon}(t)=-\boldsymbol{\phi}(t)^T\tilde{\boldsymbol{\theta}}(t) \tag{166}$$

Let $(\mathbf{A}_H,\ \mathbf{B}_H,\ \mathbf{C}_H,\ \mathbf{D}_H)$ be the state-space matrices of a minimal realization for the transfer function $H(s)$. Then the relationship (81) can be represented in a state-space form by

$$\begin{aligned}\dot{\mathbf{x}}_H(t) &= \mathbf{A}_H\mathbf{x}_H(t)+\mathbf{B}_H\boldsymbol{\epsilon}(t)\\ \boldsymbol{\nu}(t) &= \mathbf{C}_H\mathbf{x}_H(t)+\mathbf{D}_H\boldsymbol{\epsilon}(t)\end{aligned} \tag{167}$$

where $\mathbf{x}_H(t)$ is the state vector of the minimal realization. Viewing (82), $(\mathbf{A}_H,\ \mathbf{B}_H,\ \mathbf{C}_H,\ \mathbf{D}_H-\frac{\mu_H}{2})I_m$ are the state-space matrices of a minimal realization for the transfer function matrix $\mathbf{H}'(s)$. As $\mathbf{H}'(s)$ is SPR by assumption, from KYP lemma, there exist $\mathbf{P}_H>0,\ \mathbf{Q}_H>0$, and $\mathbf{L}_H$ and $\mathbf{W}_H$ such that

$$\begin{aligned}&\mathbf{A}_H^T\mathbf{P}_H+\mathbf{P}_H\mathbf{A}_H=-\mathbf{L}_H\mathbf{L}_H^T-\mathbf{Q}_H\\ &\mathbf{B}_H^T\mathbf{P}_H=\mathbf{C}_H-\mathbf{W}_H^T\mathbf{L}_H^T\\ &\mathbf{W}_H^T\mathbf{W}_H=D_H-\tfrac{\mu_H}{2}I_m+\left(D_H-\tfrac{\mu_H}{2}I_m\right)^T=\mathbf{D}_H+\mathbf{D}_H^T-\mu_H I_m\end{aligned} \tag{168}$$

Define

$$V_H(t)=\mathbf{x}_H(t)^T\mathbf{P}_H\mathbf{x}_H(t) \tag{169}$$

Then, from (167) and (168),

$$\begin{aligned}
\dot{V}_H(t) &= \dot{\mathbf{x}}_H(t)^T\mathbf{P}_H\mathbf{x}_H(t) + \mathbf{x}_H(t)^T\mathbf{P}_H\dot{\mathbf{x}}_H(t) \\
&= \left[\mathbf{A}_H\mathbf{x}_H(t) + \mathbf{B}_H\boldsymbol{\epsilon}(t)\right]^T\mathbf{P}_H\mathbf{x}_H(t) + \mathbf{x}_H(t)^T\mathbf{P}_H\left[\mathbf{A}_H\mathbf{x}_H(t) + \mathbf{B}_H\boldsymbol{\epsilon}(t)\right] \\
&= \mathbf{x}_H(t)^T\left[\mathbf{A}_H^T\mathbf{P}_H + \mathbf{P}_H\mathbf{A}_H\right]\mathbf{x}_H(t) + 2\boldsymbol{\epsilon}(t)^T\mathbf{B}_H^T\mathbf{P}_H\mathbf{x}_H(t) \\
&= \mathbf{x}_H(t)^T\left[-\mathbf{L}_H\mathbf{L}_H^T - \mathbf{Q}_H\right]\mathbf{x}_H(t) + 2\boldsymbol{\epsilon}(t)^T\left[\mathbf{C}_H - \mathbf{W}_H^T\mathbf{L}_H^T\right]\mathbf{x}_H(t) \\
&= -\mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) - \|\mathbf{L}_H^T\mathbf{x}_H(t)\|^2 \\
&\quad +2\boldsymbol{\epsilon}(t)^T\left[\boldsymbol{\nu}(t) - \mathbf{D}_H\boldsymbol{\epsilon}(t)\right] - 2\boldsymbol{\epsilon}(t)^T\mathbf{W}_H^T\mathbf{L}_H^T\mathbf{x}_H(t) \\
&= -\mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) - \|\mathbf{L}_H^T\mathbf{x}_H(t) + \mathbf{W}_H\boldsymbol{\epsilon}(t)\|^2 \\
&\quad +2\boldsymbol{\epsilon}(t)^T\boldsymbol{\nu}(t) - \boldsymbol{\epsilon}(t)^T\left[\mathbf{D}_H + \mathbf{D}_H^T - \mathbf{W}_H^T\mathbf{W}_H\right]\boldsymbol{\epsilon}(t) \\
&= -\mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) - \|\mathbf{L}_H^T\mathbf{x}_H(t) + \mathbf{W}_H\boldsymbol{\epsilon}(t)\|^2 + 2\boldsymbol{\epsilon}(t)^T\boldsymbol{\nu}(t) - \mu_H\|\boldsymbol{\epsilon}(t)\|^2 \\
&\le -\mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) + 2\boldsymbol{\epsilon}(t)^T\boldsymbol{\nu}(t) - \mu_H\|\boldsymbol{\epsilon}(t)\|^2
\end{aligned} \tag{170}$$

Thus, $\forall t$

$$2\boldsymbol{\epsilon}(t)^T\boldsymbol{\nu}(t) \ge \dot{V}_H(t) + \mu_H\|\boldsymbol{\epsilon}(t)\|^2 + \mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) \tag{171}$$

On the other hand, from (80) and (166), when the first condition in the last equation of (80) is satisfied,

$$\begin{aligned}
\dot{V}_\theta(t) &= 2\tilde{\boldsymbol{\theta}}(t)^T\boldsymbol{\Gamma}(t)^{-1}\dot{\tilde{\boldsymbol{\theta}}}(t) + \tilde{\boldsymbol{\theta}}(t)^T\tfrac{d\boldsymbol{\Gamma}(t)^{-1}}{dt}\tilde{\boldsymbol{\theta}}(t) \\
&= 2\tilde{\boldsymbol{\theta}}(t)^T\boldsymbol{\phi}(t)\boldsymbol{\nu}(t) + \tilde{\boldsymbol{\theta}}(t)^T\left[-\mu_1(t)\boldsymbol{\Gamma}(t)^{-1} + \mu_2(t)\boldsymbol{\phi}(t)\boldsymbol{\phi}(t)^T\right]\tilde{\boldsymbol{\theta}}(t) \\
&= -2\boldsymbol{\epsilon}(t)^T\boldsymbol{\nu}(t) - \mu_1(t)\tilde{\boldsymbol{\theta}}(t)^T\boldsymbol{\Gamma}(t)^{-1}\tilde{\boldsymbol{\theta}}(t) + \mu_2(t)\|\boldsymbol{\epsilon}(t)\|^2
\end{aligned} \tag{172}$$

Substituting (171) into (172),

$$\dot{V}_\theta(t) \le -\dot{V}_H(t) - \mu_\epsilon\|\boldsymbol{\epsilon}(t)\|^2 - \mathbf{x}_H(t)^T\mathbf{Q}_H\mathbf{x}_H(t) \tag{173}$$

where $\mu_\epsilon = \mu_{2H} - \mu_{2M} > 0$. Thus, $\forall t$,

$$\begin{aligned}
\mu_\epsilon\int_0^t\|\boldsymbol{\epsilon}(\tau)\|^2d\tau + \lambda_{min}(Q_H)\int_0^t\|\mathbf{x}_H(\tau)\|^2d\tau &\le -\int_0^t[\dot{V}_\theta(\tau) + \dot{V}_H(\tau)]d\tau \\
= V_\theta(0) - V_\theta(t) + V_H(0) - V_H(t) &\le V_\theta(0) + V_H(0)
\end{aligned} \tag{174}$$

So $\boldsymbol{\epsilon} \in \mathcal{L}_2^m$ and $\mathbf{x}_H \in \mathcal{L}_2^{n_H}$. Noting (167), $\boldsymbol{\nu} \in \mathcal{L}_2^m$ as well, which proves P2 of the theorem. It is also easy to see from (174) that $V_\theta(t) \le V_\theta(0) + V_H(0)$. Thus $V_\theta(t) \in \mathcal{L}_\infty$. Due to the use of co-variance limiting in (80), $\boldsymbol{\Gamma}(t) \le \rho_M I_p$. So $V_\theta(t) \in \mathcal{L}_\infty$ also implies $\tilde{\boldsymbol{\theta}} \in \mathcal{L}_\infty^p$, which proves P1 of the theorem. When $\boldsymbol{\phi} \in \mathcal{L}_\infty^{p\times m}$, as $\boldsymbol{\Gamma}(t) \le \rho_M I_p$, $\boldsymbol{\nu} \in \mathcal{L}_2^m$ also implies $\tilde{\boldsymbol{\theta}} \in \mathcal{L}_2^p$, which leads to P3 of the theorem. ♢

Proof. **Proof of Theorem 4**
Let

$$A(q) = a_0q^{n-1} + a_1q^{n-2} + \cdots + a_{n-1} \tag{175}$$

Then, $A(q)$ being all nonzero polynomials of degree $n-1$ or less is the same as $\mathbf{a} = \left[a_0\ a_1\ \cdots\ a_{n-1}\right] \in R^n$ in (175) being arbitrary. Since

$$
\begin{aligned}
U &= \lim_{t\to\infty} \tfrac{1}{t} \sum_{k=1}^{t} \left[a_0 u(k+n-1) + a_1 u(k+n-2) + \cdots + a_{n-1} u(k)\right]^2 \\
&= \lim_{t\to\infty} \tfrac{1}{t} \sum_{k=1}^{t} \mathbf{a}^T \begin{bmatrix} u(k+n-1) \\ \vdots \\ u(k) \end{bmatrix} \begin{bmatrix} u(k+n-1) \cdots u(k) \end{bmatrix} \mathbf{a} \\
&= \mathbf{a}^T \left\{ \lim_{t\to\infty} \tfrac{1}{t} \sum_{k=1}^{t} \begin{bmatrix} u(k+n-1) \\ \vdots \\ u(k) \end{bmatrix} \begin{bmatrix} u(k+n-1) \cdots u(k) \end{bmatrix} \right\} \mathbf{a} \\
&= \mathbf{a}^T \mathbf{C}_n \mathbf{a}
\end{aligned}
\tag{176}
$$

then, $U > 0$ for all $\mathbf{a}$ is the same as that $\mathbf{C}_n$ is a positive definite matrix or u is weakly persistently exciting of order n, which completes the proof. ♢

Proof. **Proof of Theorem 5**
Let $\phi_u(\omega)$ be the spectral density function of $\{u(t)\}$ and we will use contradiction to prove the theorem. So let us assume that u is NOT weakly persistently exciting of order n. From Theorem 4, $\exists \mathbf{a} \neq 0 \in R^n$ such that $U = 0$. Let $A(q)$ be the non-zero $(n-1)$-th polynomial corresponding to such an $\mathbf{a}$. On the other hand, from Parseval's theorem[4],

$$
U = \lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} \left(A(q)u(k)\right)^2 = R_{yy}(0) = \frac{1}{2\pi} \int_{-\pi}^{\pi} \left|A(e^{i\omega})\right|^2 \phi_u(\omega) d\omega \tag{177}
$$

Since $U = 0$, the right hand side of (177) is zero as well. Thus,

$$
\left|A(e^{i\omega})\right|^2 \phi_u(\omega) = 0, \quad \forall \omega \in \left[-\pi, \pi\right] \tag{178}
$$

Since $\phi_u(\omega)$ is non-zero at least at n points $\omega_1, \omega_2, \cdots \omega_n$, we have

$$
A(e^{i\omega_k}) = 0, \quad k = 1, 2, \cdots, n \tag{179}
$$

which leads to the conclusion that $A(z) = 0$ has n solutions. But since $A(z)$ is a polynomial of order $n-1$ only and $A(z) = 0$ has at most $n-1$ solutions, a contradiction results and the theorem is thus proven. ♢

Proof. **Proof of Theorem 6**
To prove (a), let $B_c(q)$ be any polynomial of degree $n-m-1$. Then, the degree of polynomial $B_c(q)B(q)$ will be less or equal to $n-1$. Since $u(t)$ is PE of order n, from Theorem 4,

$$
U = \lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} \left(B_c(q)B(q)u(k)\right)^2 > 0 \tag{180}
$$

[4] For an LTI system with a TF of $A(z)$ and an input of spectrum $\phi_u(\omega)$, the spectral density function of the output y is $\phi_y(\omega) = \left|A(e^{i\omega})\right|^2 \phi_u(\omega)$

This shows that $\lim_{t\to\infty} \frac{1}{t} \sum_{k=1}^{t} (B_c(q)w(k))^2 > 0,\ \forall B_c(q)$. By Theorem 4, w is PE of order $n-m$, which completes the proof of (a).

The following is to prove (b). Since $u(t)$ is PE of order n, from Theorem 5, its spectrum $\phi_u(\omega)$ is non-zero at n points. As $\frac{1}{A(e^{i\omega})} \neq 0,\ \forall\omega$, the spectrum of w, $\phi_w(\omega) = \left|\frac{1}{A(e^{i\omega})}\right|^2 \phi_u(\omega)$, is non-zero at n points as well. Thus, by Theorem 5, w is PE of order n. This completes the proof. ♢

Proof. **Proof of Theorem 7**
For notational simplicity, let $v(t) = \boldsymbol{\varphi}(t)^T\boldsymbol{\theta}$. Noting that the actual output $y(t)$ is generated by (137) with $e(t) = 0$, straightforward checking shows that $\forall\theta \in \mathcal{R}^{n_a+n_b+1}$,

$$\begin{aligned} v(t) &= [q^{n_a} - A(q)]\, q^{-n_a} y(t) + B(q) q^{-n_a} u(t) \\ &= \left\{ \underbrace{[q^{n_a} - A(q)]B^0(q) + A^0(q)B(q)}_{L(q)} \right\} \underbrace{\frac{1}{A^0(q)} u(t-n_a)}_{w(t)} \end{aligned} \tag{181}$$

From Theorem 6, the signal $w(t)$ in (181) is PE of order $p = n_a + n_b + 1$. As $n_a = n_{A^0}$ and $n_b \geq n_{B^0}$, the polynomial $L(q)$ in (181) at most has an order of $n_a + n_b$. Furthermore, $L(q) \neq 0$ as $L(q) = 0$ would lead to

$$\frac{B^0(q)}{A^0(q)} = \frac{B(q)}{q^{n_a} - A(q)} \tag{182}$$

which contradicts with the co-prime assumption of $B^0(q)$ and $A^0(q)$ when noting the order of $q^{n_a} - A(q)$ is $n_a - 1 < n_{A^0}$. Thus, By Theorem (6), $v(t)$ is PE order of at least 1. Thus, by Theorem (4),

$$\boldsymbol{\theta}^T \lim_{t\to\infty} \tfrac{1}{t} \boldsymbol{\Phi}(t)^T \boldsymbol{\Phi}(t) \boldsymbol{\theta} = \lim_{t\to\infty} \tfrac{1}{t} \sum_{k=1}^{t} (v(k))^2 > 0 \tag{183}$$

which completes the proof. ♢

Proof. **Proof of Theorem 8**
Without loss of generality, assume $t_0 = 0$ in (141). From the first inequality of the PE condition (141),

$$\rho_1 \|\tilde{\boldsymbol{\theta}}(jl)\|^2 \leq \tilde{\boldsymbol{\theta}}(jl)^T \left[\sum_{i=jl+1}^{jl+l} \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \right] \tilde{\boldsymbol{\theta}}(jl), \quad \forall j \tag{184}$$

Thus, $\forall k \geq 0$,

$$\rho_1 \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jl)\|^2 \leq \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \tilde{\boldsymbol{\theta}}(jl)^T \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \tilde{\boldsymbol{\theta}}(jl) \tag{185}$$

By the definition of $\epsilon(t)$ in (58),

$$\sum_{i=1}^{(k+1)l} \epsilon(i)^2 = \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \epsilon(i)^2 = \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \tilde{\boldsymbol{\theta}}(i)^T \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \tilde{\boldsymbol{\theta}}(i) \tag{186}$$

Noting that the second inequality of the PE condition (141) implies that $\forall t,\ \|\boldsymbol{\phi}(t)\| \leq \rho_\phi$ for some $\rho_\phi > 0$, the difference between the right hand sides of (185) and (186) is bounded above by

$$\begin{aligned}
\rho_1 \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jl)\|^2 - \sum_{i=1}^{(k+1)l} \epsilon(i)^2 &= \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \Big\{ -2\tilde{\boldsymbol{\theta}}(jl)^T \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \left[\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\right] \\
&\qquad + \left[\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\right]^T \boldsymbol{\phi}(i)\boldsymbol{\phi}(i)^T \left[\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\right] \Big\} \\
&\leq \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \left\{ 2\rho_\phi^2 \|\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\| \|\tilde{\boldsymbol{\theta}}(jl)\| + \rho_\phi^2 \|\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\|^2 \right\} \\
&\leq \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \left\{ \tfrac{\rho_1}{2l} \|\tilde{\boldsymbol{\theta}}(jl)\|^2 + \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \|\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(jl)\|^2 \right\} \\
&\leq \tfrac{\rho_1}{2} \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jl)\|^2 + \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \left[\sum_{p=jl+1}^{i} \|\tilde{\boldsymbol{\theta}}(p) - \tilde{\boldsymbol{\theta}}(p-1)\| \right]^2
\end{aligned} \tag{187}$$

Thus,

$$\begin{aligned}
\sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jl)\|^2 &\leq \tfrac{2}{\rho_1} \left\{ \sum_{i=1}^{(k+1)l} \epsilon(i)^2 + \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} \left[\sum_{p=jl+1}^{(j+1)l} \|\tilde{\boldsymbol{\theta}}(p) - \tilde{\boldsymbol{\theta}}(p-1)\| \right]^2 \right\} \\
&\leq \tfrac{2}{\rho_1} \left\{ \sum_{i=1}^{(k+1)l} \epsilon(i)^2 + \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \sum_{j=0}^{k} \sum_{i=jl+1}^{(j+1)l} l \sum_{p=jl+1}^{(j+1)l} \|\tilde{\boldsymbol{\theta}}(p) - \tilde{\boldsymbol{\theta}}(p-1)\|^2 \right\} \\
&\leq \tfrac{2}{\rho_1} \left\{ \sum_{i=1}^{(k+1)l} \epsilon(i)^2 + l^2 \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \sum_{j=0}^{k} \sum_{p=jl+1}^{(j+1)l} \|\tilde{\boldsymbol{\theta}}(p) - \tilde{\boldsymbol{\theta}}(p-1)\|^2 \right\} \\
&\leq \tfrac{2}{\rho_1} \left\{ \sum_{i=1}^{(k+1)l} \epsilon(i)^2 + l^2 \left(\tfrac{2l\rho_\phi^4}{\rho_1} + \rho_\phi^2\right) \sum_{i=1}^{(k+1)l} \|\tilde{\boldsymbol{\theta}}(i) - \tilde{\boldsymbol{\theta}}(i-1)\|^2 \right\}
\end{aligned} \tag{188}$$

On the other hand,

$$\begin{aligned}\sum_{i=1}^{(k+1)l}\|\tilde{\boldsymbol{\theta}}(i)\|^2 &= \sum_{j=0}^{k}\sum_{i=jl+1}^{(j+1)l}\left\|\tilde{\boldsymbol{\theta}}(jl)+\sum_{p=jl+1}^{i}\left\{\tilde{\boldsymbol{\theta}}(p)-\tilde{\boldsymbol{\theta}}(p-1)\right\}\right\|^2\\ &\le \sum_{j=0}^{k}\sum_{i=jl+1}^{(j+1)l}\left\{\|\tilde{\boldsymbol{\theta}}(jl)\|+\sum_{p=jl+1}^{(j+1)l}\|\tilde{\boldsymbol{\theta}}(p)-\tilde{\boldsymbol{\theta}}(p-1)\|\right\}^2\\ &\le (l+1)\sum_{j=0}^{k}\sum_{i=jl+1}^{(j+1)l}\left\{\|\tilde{\boldsymbol{\theta}}(jl)\|^2+\sum_{p=jl+1}^{(j+1)l}\|\tilde{\boldsymbol{\theta}}(p)-\tilde{\boldsymbol{\theta}}(p-1)\|^2\right\}\\ &= (l+1)l\left\{\sum_{j=0}^{k}\|\tilde{\boldsymbol{\theta}}(jl)\|^2+\sum_{j=0}^{k}\sum_{p=jl+1}^{(j+1)l}\|\tilde{\boldsymbol{\theta}}(p)-\tilde{\boldsymbol{\theta}}(p-1)\|^2\right\}\\ &= (l+1)l\left\{\sum_{j=0}^{k}\|\tilde{\boldsymbol{\theta}}(jl)\|^2+\sum_{i=1}^{(k+1)l}\|\tilde{\boldsymbol{\theta}}(p)-\tilde{\boldsymbol{\theta}}(p-1)\|^2\right\}\end{aligned} \tag{189}$$

From (62) and (159), it is clear that $\tilde{\boldsymbol{\theta}}(t)-\tilde{\boldsymbol{\theta}}(t-1)=\hat{\boldsymbol{\theta}}(t)-\hat{\boldsymbol{\theta}}(t-1)\in l_2$. As $\epsilon\in l_2$ as well, the right hand side of (188) has a finite limit as $k\to\infty$. Thus $\lim_{k\to\infty}\sum_{j=0}^{k}\|\tilde{\boldsymbol{\theta}}(jl)\|^2<\infty$, and the right hand side of (189) has a finite limit when $k\to\infty$. So $\tilde{\boldsymbol{\theta}}\in l_2$ and, as such, $\tilde{\boldsymbol{\theta}}(t)\to 0$ as $t\to\infty$, which completes the proof. ♢

Proof. **Proof of Theorem 9**
Without loss of generality, assume $t_0=0$ in (142). Then the PE condition (142) implies that

$$\rho T\|\tilde{\boldsymbol{\theta}}(jT)\|^2\le\tilde{\boldsymbol{\theta}}(jT)^T\left[\int_{jT}^{(j+1)T}\boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T d\tau\right]\tilde{\boldsymbol{\theta}}(jT),\quad\forall j\ge 0 \tag{190}$$

Thus, $\forall k\ge 0$,

$$\rho T\sum_{j=0}^{k}\|\tilde{\boldsymbol{\theta}}(jT)\|^2\le\sum_{j=0}^{k}\int_{jT}^{(j+1)T}\left[\tilde{\boldsymbol{\theta}}(jT)^T\boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T\tilde{\boldsymbol{\theta}}(jT)\right]d\tau \tag{191}$$

By the definition of $\epsilon(t)$ in (81),

$$\int_0^{(k+1)T}\epsilon(\tau)^2 d\tau=\sum_{j=0}^{k}\int_{jT}^{(j+1)T}\epsilon(\tau)^2 d\tau=\sum_{j=0}^{k}\int_{jT}^{(j+1)T}\left[\tilde{\boldsymbol{\theta}}(\tau)^T\boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T\tilde{\boldsymbol{\theta}}(\tau)\right]d\tau \tag{192}$$

The difference between the right hand sides of (191) and (192) is bounded above by

$$\begin{aligned}
\rho T \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 - \int_0^{(k+1)T} \epsilon(\tau)^2 d\tau &= \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\{ -2\tilde{\boldsymbol{\theta}}(jT)^T \boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T \\
&\quad \Big[\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\Big] + \Big[\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\Big]^T \boldsymbol{\phi}(\tau)\boldsymbol{\phi}(\tau)^T \Big[\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\Big] \Big\} d\tau \\
&\le \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\{ 2\rho_\phi^2 \|\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\| \|\tilde{\boldsymbol{\theta}}(jT)\| + \rho_\phi^2 \|\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\|^2 \Big\} d\tau \\
&\le \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\{ \tfrac{\rho}{2} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + \Big(\tfrac{2\rho_\phi^4}{\rho} + \rho_\phi^2\Big) \|\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\|^2 \Big\} d\tau \\
&\le \tfrac{\rho T}{2} \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + \Big(\tfrac{2\rho_\phi^4}{\rho} + \rho_\phi^2\Big) \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big[\| \int_{jT}^{\tau} \dot{\tilde{\boldsymbol{\theta}}}(s) ds \| \Big]^2 d\tau
\end{aligned} \tag{193}$$

Thus,

$$\begin{aligned}
\sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 &\le \tfrac{2}{\rho T} \Bigg\{ \int_0^{(k+1)T} \epsilon(\tau)^2 d\tau + \Big(\tfrac{2\rho_\phi^4}{\rho} + \rho_\phi^2\Big) \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big[\int_{jT}^{\tau} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\| ds \Big]^2 d\tau \Bigg\} \\
&\le \tfrac{2}{\rho T} \Bigg\{ \int_0^{(k+1)T} \epsilon(\tau)^2 d\tau + \Big(\tfrac{2\rho_\phi^4}{\rho} + \rho_\phi^2\Big) T \sum_{j=0}^{k} \Big[\int_{jT}^{(j+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\| ds \Big]^2 \Bigg\}
\end{aligned} \tag{194}$$

Applying the Cauchy-Schwartz inequality,

$$\begin{aligned}
\Big[\int_{jT}^{(j+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\| ds \Big]^2 &\le \Bigg[\sqrt{\int_{jT}^{(j+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\|^2 ds} \sqrt{\int_{jT}^{(j+1)T} 1^2 ds} \Bigg]^2 \\
&= T \int_{jT}^{(j+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\|^2 ds
\end{aligned} \tag{195}$$

Substituting (195) into (195),

$$\sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 \le \tfrac{2}{\rho T} \Big\{ \int_0^{(k+1)T} \epsilon(\tau)^2 d\tau + \Big(\tfrac{2\rho_\phi^4}{\rho} + \rho_\phi^2\Big) T \int_0^{(k+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\|^2 ds \Big\} \tag{196}$$

which shows that $\sum_{j=0}^{\infty} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 < \infty$ by letting $k \to \infty$ as $\epsilon \in \mathcal{L}$ and $\dot{\tilde{\boldsymbol{\theta}}} \in \mathcal{L}_2^p$ from P2 and P3 of Theorem 3. On the other hand,

$$\begin{aligned}
\int_0^{(k+1)T} \|\tilde{\boldsymbol{\theta}}(\tau)\|^2 d\tau &= \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\| \tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT) + \tilde{\boldsymbol{\theta}}(jT) \Big\|^2 d\tau \\
&\le \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\{ \|\tilde{\boldsymbol{\theta}}(jT)\| + \|\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\| \Big\}^2 d\tau \\
&\le 2 \sum_{j=0}^{k} \int_{jT}^{(j+1)T} \Big\{ \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + \|\tilde{\boldsymbol{\theta}}(\tau) - \tilde{\boldsymbol{\theta}}(jT)\|^2 \Big\} d\tau \\
&= 2 \sum_{j=0}^{k} \Bigg\{ T \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + \int_{jT}^{(j+1)T} \Big[\| \int_{jT}^{\tau} \dot{\tilde{\boldsymbol{\theta}}}(s) ds \| \Big]^2 d\tau \Bigg\} \\
&\le 2T \sum_{j=0}^{k} \Bigg\{ \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + \Big[\int_{jT}^{(j+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\| ds \Big]^2 \Bigg\}
\end{aligned} \tag{197}$$

Substituting (195) into (195)

$$\int_0^{(k+1)T} \|\tilde{\boldsymbol{\theta}}(\tau)\|^2 d\tau =\leq 2T \sum_{j=0}^{k} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 + 2T^2 \int_0^{(k+1)T} \|\dot{\tilde{\boldsymbol{\theta}}}(s)\|^2 ds \qquad (198)$$

which shows that $\tilde{\boldsymbol{\theta}} \in \mathcal{L}_2^p$ by letting $k \to \infty$ as $\sum_{j=0}^{\infty} \|\tilde{\boldsymbol{\theta}}(jT)\|^2 < \infty$ and $\dot{\tilde{\boldsymbol{\theta}}} \in \mathcal{L}_2^p$. Since $\dot{\tilde{\boldsymbol{\theta}}} \in \mathcal{L}_\infty^p$ as well, $\tilde{\boldsymbol{\theta}}(t)$ is a uniformly continuous function of time. As such, by Barbalat's lemma [7], $\tilde{\boldsymbol{\theta}}(t) \to 0$ as $t \to \infty$, which completes the proof. $\diamondsuit$